

A collision-free motion planning method for cable-drive redundant manipulators with deep reinforcement learning-based expert guidance and long short-term memory

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ABSTRACT

To address the challenge where traditional control methods struggle to balance real-time performance and robustness for cable-driven redundant manipulators due to strong nonlinearity, this paper proposes an end-to-end deep reinforcement learning control framework integrating expert guidance, curriculum learning, and Long Short-Term Memory (LSTM) networks. By utilizing a hierarchical expert demonstration generation mechanism and a three-stage curriculum learning strategy, the framework maps historical state sequences to control actions, significantly improving sample efficiency and training stability. Experimental results indicate that the proposed SAC-10%-LSTM model achieves optimal comprehensive performance in complex multi-obstacle scenarios, reaching an average cumulative reward of 33,400.22 and an average final distance of 0.033 m with zero collisions. Compared to the GRU model, which reduces decision steps by 24.7% but yields a slightly lower average return of 29,945.34, the proposed model improves returns by approximately 10.3% and effectively overcomes the low rewards of 7290.89 and high collision rates caused by the short-sightedness of the TCN model. Parameter sensitivity analysis confirms that an input sequence length of 30 is optimal, achieving a 100% task success rate and a final distance of 0.0330 m within only 2.00 h of training, outperforming non-convergent long sequences larger than 70 and failing short sequences of 10. Furthermore, across nine generalization scenarios with one to three obstacles, the final distance stabilizes between 0.028 m and 0.040 m, effectively resolving rigid constraint jitter, geometric deadlocks, and local minima issues inherent in traditional APF methods, thereby providing an efficient paradigm for the intelligent control of high-degree-of-freedom underactuated systems.

1. Introduction

Robots have been widely applied in industries such as manufacturing (Kadri, Selouani, Ghribi, Ghali, & Mekhoukh, 2025), healthcare (Shaw & Chen, 2025), and space exploration (Yin, Liu, Rui, & Yu, 2025). Conventional rigid robotic arms, characterized by complex, stiff link structures and heavy actuators that inevitably result in large mass and

high inertia, suffer from physical constraints that severely hinder their flexibility and deployment capabilities. Cable-driven Redundant Manipulators, as an advanced rigid-flexible hybrid configuration, have gained extensive applications in medical devices, operations in confined spaces, and space exploration due to their lightweight nature and high compliance (Chen et al., 2025; Li et al., 2022; Li et al., 2024; Lin and Zhou, 2022; Tanaka and Hamaya, 2023; Xu et al., 2020; Zhang et al.,

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2025). Owing to the presence of nonlinear factors in cable-driven systems, current research on cable-driven robotic arms primarily focuses on kinematic and dynamic modeling, as well as control strategy development.

Despite the operational advantages of cable-driven Redundant Manipulators, their deployment is complicated by inherent nonlinear dynamics, including friction and hysteresis. The challenge of establishing accurate analytical models has necessitated the development of diverse approximation techniques. Inspired by biological mechanisms such as octopus tentacles, recent research has established analytical static models that account for transverse cable deformation (Qi, Mei, Chen, Li, & Tan, 2024); while neural networks optimized via the Whale Optimization Algorithm have been employed to approximate dynamic behaviors (Zhou et al., 2024). Addressing the complexities of hyper-redundancy, researchers have improved Denavit-Hartenberg parameters to construct Jacobian matrices (Huang, Shen, Mei, & Chen, 2025) or adopted Piecewise Constant Curvature frameworks (Lai, Huang, Lu, Zhao, & Chu, 2022). In parallel, kinematic simplifications have led to equations replacing sub-joint variables (Liu, Xu, Yang, & Li, 2021). and equivalent Jacobian matrices for segmented rotation axes (Hu et al., 2019). Regarding static analysis, comprehensive models incorporating gravity, external loads, and friction have been established utilizing Newton-Euler methods (Mu et al., 2025).

Building upon these modeling foundations, traditional control strategies have been extensively explored. Sliding Mode Control based on Time Delay Estimation has been investigated to compensate for unknown system dynamics (Wang, Jiang, Yan, & Chen, 2017); while co-ordinated adaptive impedance control laws have been developed to manage compliant interactions (Ma, Cheneler, He, Yuan, & Bian, 2025). Further methodological advancements include two-layer geometric iteration techniques for inverse kinematics (Liu, Yang, Xu, Mylonas, & Liang, 2022). and self-calibration methods that enhance motion accuracy through optimized tension control (Chen, Li, Wu, Liu, & Peng, 2024).

The theoretical foundation of the aforementioned control strategies relies heavily on deterministic analytical models, such as precise Jacobian matrices and fixed dynamic parameters. While effective in structured settings, these model-based methods encounter a fundamental bottleneck in unstructured environments. The randomness of target positions and the dynamic evolution of obstacles introduce time-varying non-linearities and unpredictability that pre-computed kinematic models cannot accommodate, leading to severe 'model mismatch' and performance degradation. In contrast, the proposed method leverages Deep Reinforcement Learning to learn an end-to-end control policy without explicit physical modeling. Furthermore, by integrating LSTM, it theoretically addresses the non-Markovian characteristics of cable dynamics, such as hysteresis and time-delayed deformation, thereby overcoming the adaptability limitations of traditional robust control.

Artificial Intelligence (AI) has witnessed exponential growth, fundamentally reshaping various scientific and engineering domains. Notably, foundation models such as Large Language Models (LLMs) have demonstrated exceptional capabilities in complex reasoning and interdisciplinary applications, providing new paradigms for intelligent systems (Guo et al., 2025). In the specific context of robotic control, AI algorithms are increasingly introduced to transcend the limitations of rigid analytical models. While traditional AI paradigms like supervised learning have achieved success in perception and classification tasks, they inherently suffer from a heavy reliance on high-quality labeled datasets, which are often scarce in robotic manipulation. Furthermore, these static learning approaches lack the mechanism to actively interact with the environment or adapt to temporal dynamic variations, rendering them insufficient for continuous control tasks.

Deep reinforcement learning (DRL) integrates deep learning with reinforcement learning to overcome existing limitations and facilitate autonomous decision-making in high-dimensional spaces by enabling an agent to interact with the environment through trial-and-error learning

and process high-dimensional sensory inputs via deep neural networks.

As shown in Table 1, current robot control methods based on Deep Reinforcement Learning have achieved some progress. Frameworks integrating forward kinematics (Chen, Su, Ni, & Li, 2024) and those leveraging expert demonstrations (Ramirez & Yu, 2023) have notably enhanced control performance. To address obstacle avoidance in redundant systems, approaches combining RL with null-space motion (Huang et al., 2023) or employing curriculum learning for optimal path tracking (Chivkula, Rodwell, & Tallapragada, 2022) have been proposed. Other significant contributions include decoupling tasks via gradient projection (Hua, Wang, Xu, & Chen, 2021); offline algorithms incorporating behavior cloning regularization (Ma et al., 2025); and hybrid planners merging artificial potential fields with Soft Actor-Critic algorithms (Bai, Zhang, Guo, & Yue, 2023). Although specific studies have attempted to incorporate Long Short-Term Memory (LSTM) networks to mitigate dynamic uncertainties (Zhou et al., 2024) existing DRL frameworks often struggle to fully address the non-Markovian characteristics, specifically hysteresis and time-delayed deformation, inherent to cable-driven mechanisms in highly unstructured environments.

Deep reinforcement learning (DRL) has been widely applied in complex robotic control tasks, enabling autonomous decision-making in high-dimensional state spaces. However, when it comes to real-robot applications in unstructured environments, DRL still faces numerous key challenges. Traditional deep reinforcement learning algorithms typically learn from scratch. Due to the presence of ineffective or hazardous actions, they not only suffer from slow convergence but also experience frequent training failures. Although previous studies have attempted to accelerate the training process by integrating expert experience through imitation learning, these methods often rely on demonstration data from real robots or manually designed reward functions, making it difficult to obtain high-quality and diverse expert data.

This paper proposes a Deep Reinforcement Learning (DRL) control framework. It synergistically integrates expert guidance, curriculum learning, and LSTM networks to enhance efficiency and robustness.

- (1) A hierarchical expert demonstration mechanism utilizing a lightweight kinematic model is introduced to incorporate prior knowledge, thereby accelerating exploration efficiency.

Table 1
Comparison of deep reinforcement learning.

Author	Framework	Key Improvement	Advantages	Limitations
Chen et al. (2024)	DRL + FK	Incorporates kinematic constraints into RL training	Improve convergence speed	Rely on FK accuracyAnd Single experimental scenario
Zhou et al. (2024)	Improved TD3 + LSTM + WOA	LSTM captures historical → current policy relationship	No need to rely on complex physical models	Time-consuming data collection
Hua et al. (2021)	RL + Gradient Projection	Task decomposition	Excellent task decoupling	Limited generalization ability
Ramirez and Yu (2023)	RL + Expert Demonstrations	Uses expert demonstration data to estimate expert policies and guide RL	Faster convergence, reduced exploration cost	Strongly depends on high-quality expert data
Bai et al. (2023)	SAC + APF	SAC for global planning, APF for local guidance	Combines global optimality with local reactivity,	Algorithmic complexity, high parameter tuning cost

- (2) An LSTM-based temporal feature extraction module is integrated into the Soft Actor-Critic (SAC) framework to theoretically address non-Markovian dynamics, specifically hysteresis and time-delayed deformation.
- (3) A three-stage curriculum learning strategy is implemented to decompose high-dimensional manipulation tasks, ensuring stable policy convergence and adaptability in complex environments.

The remainder of this paper is organized as follows: [Section 2](#) formulates the motion planning problem as a partially observable Markov decision process and details the proposed framework—including the hierarchical expert demonstration generation mechanism, dual experience replay with reward shaping, three-stage curriculum learning strategy, and the Long Short-Term Memory (LSTM)-augmented Soft Actor-Critic architecture. [Section 3](#) describes the experimental setup, covering the manipulator model, training protocol, and evaluation metrics. [Section 4](#) validates the effectiveness of individual components and the overall framework through comparative experiments. [Section 5](#) concludes the paper and discusses future research directions.

2. Methodology

2.1. Hybrid planning framework for cable-driven manipulators in complex unstructured environments

Addressing the challenges of high exploration costs and susceptibility to local optima in motion planning for Cable-drive Redundant Manipulators operating in complex, unstructured environments, and proposes a hybrid framework that integrates traditional path planning methods with deep reinforcement learning.

The research subject is a high-degree-of-freedom flexible Cable-drive

Redundant Manipulator, whose core structure consists of four rigid links connected in series through universal joints, driven cooperatively by 12 cables. By precisely adjusting the rotational speeds of 12 motors to change cable lengths, control over the overall configuration of the manipulator and the position of the end-effector is achieved. This cable-driven structure endows the Cable-drive Redundant Manipulator with excellent compliance and a large workspace, while simultaneously introducing complex control challenges, including strong nonlinear dynamics and multivariable coupling effects, rendering traditional motion planning methods difficult to apply effectively.

The overall architecture is illustrated in [Fig. 1](#). Specifically, the framework employs the Artificial Potential Field (APF) method to generate trajectories, which are used as expert demonstration samples. A dual experience replay buffer is constructed within the deep reinforcement learning component to incorporate these expert demonstrations into the learning process. Subsequently, the LSTM-SAC algorithm is employed for training.

In order to overcome the challenges associated with directly solving inverse kinematics within the true dynamic model and to mitigate blind exploration during the initial phases of reinforcement learning, this paper employs a simplified rigid-link model integrated with traditional path planning algorithms to generate a substantial volume of feasible, collision-free trajectories. These generated trajectories constitute structured expert experience and are pre-loaded into the Expert experience replay buffer, as illustrated in the bottom-left corner of [Fig. 1](#). Such an initialization strategy ensures the acquisition of high-quality reference data at the training onset, thereby effectively guiding agent exploration within the high-dimensional action space and accelerating policy convergence.

By substituting the conventional MLP architecture with a Long Short-Term Memory (LSTM) network as a state encoder within the SAC

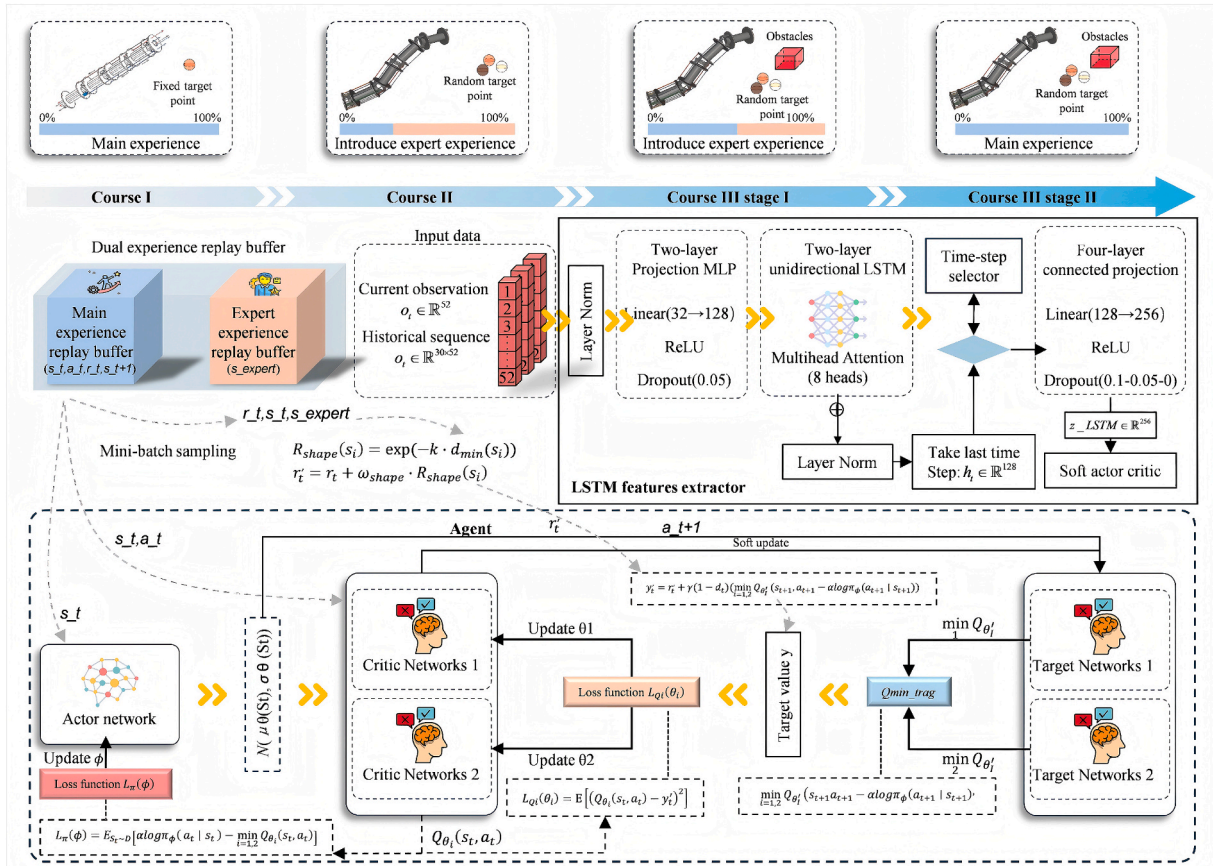


Fig. 1. Algorithm framework.

algorithm, the framework effectively handles the partial observability inherent in the dynamics of cable-driven manipulators. As illustrated in the LSTM Features Extractor module in Fig. 1, the system accepts historical sequence data O_t as input.

The input data first undergoes Layer Normalization and is subsequently mapped into a latent space via a linear layer incorporating ReLU activation and a Dropout mechanism. Subsequently, the data flows through a two-layer unidirectional LSTM and captures temporal dependencies facilitated by an 8-head multi-head attention mechanism. Finally, a Time-step selector extracts the critical hidden state, which is then processed by a four-layer fully connected projection to output a 256-dimensional feature vector Z_{LSTM} , thereby empowering the policy to leverage historical information for decision-making within dynamic environments.

The framework designs a dual experience replay buffer architecture that concurrently maintains a Main Experience Replay Buffer for data derived from autonomous interactions and an Expert Experience Replay Buffer for expert trajectories. In conjunction with a reward shaping mechanism based on state similarity, during the mini-batch sampling phase, the system computes the minimum Euclidean distance $d_{min}(s_t)$ between the current state and the expert trajectories, transforming this metric into a guidance reward. The integration of the guidance reward with the original environmental reward r_t yields a composite signal r'_t , utilized within the Actor and Critic update loops, effectively steering the policy toward safe obstacle-avoidance regions in the high-dimensional action space while simultaneously preserving sufficient environmental exploration.

Within the overall algorithmic framework, the training process adopts a three-stage curriculum learning strategy as illustrated in the top timeline of Fig. 1, characterized by a stepwise escalation in task complexity. The training regimen initiates with fixed point-to-point tasks in an obstacle-free environment (Course I), relying exclusively on main experience for fundamental kinematic acquisition; subsequently, it progresses to random target tracking (Course II), where expert experience is introduced to augment generalization capabilities.

The final phase (Course III) is conducted in complex scenarios containing dynamic obstacles and implements two parallel strategies to investigate distinct learning mechanisms: Strategy 1 continues to employ expert experience for guidance, while Strategy 2 transitions the network architecture to an LSTM-based structure, leveraging its memory mechanism to temporally encode historical observation sequences to effectively address the partial observability problem in unstructured environments, thereby facilitating the agent's acquisition of complex dynamic obstacle avoidance capabilities.

2.2. Course learning

As illustrated in Fig. 2, a hybrid three-stage curriculum learning strategy is implemented to guide the agent from basic motion control to obstacle avoidance in complex scenarios by progressively increasing task difficulty.

In the first stage, training focuses on the construction of basic mappings. The agent interacts autonomously in an obstacle-free environment with static targets using only standard reinforcement learning, without incorporating any expert demonstrations. This process aims to enable the agent to grasp the kinematics and dynamics of the robotic arm, establishing a precise mapping from configuration space to task space, with the resulting model serving as effective pre-trained weights for the subsequent stages.

The second stage emphasizes enhancing the policy's generalization capabilities regarding dynamic targets. The system introduces random target points within an obstacle-free environment, initialized based on the model from the previous stage. By employing a reward shaping mechanism, the state distribution of expert demonstrations is integrated into the algorithm's update process. This guides the agent to learn smoother motion trajectories while adapting to arbitrary targets.

The third stage advances to a complex environment containing obstacles. Building upon the second-stage model, two strategies are explored in parallel. One continues the hybrid learning mode, utilizing expert obstacle avoidance trajectories for explicit guidance to accelerate learning. The other strategy discards expert data and transitions the

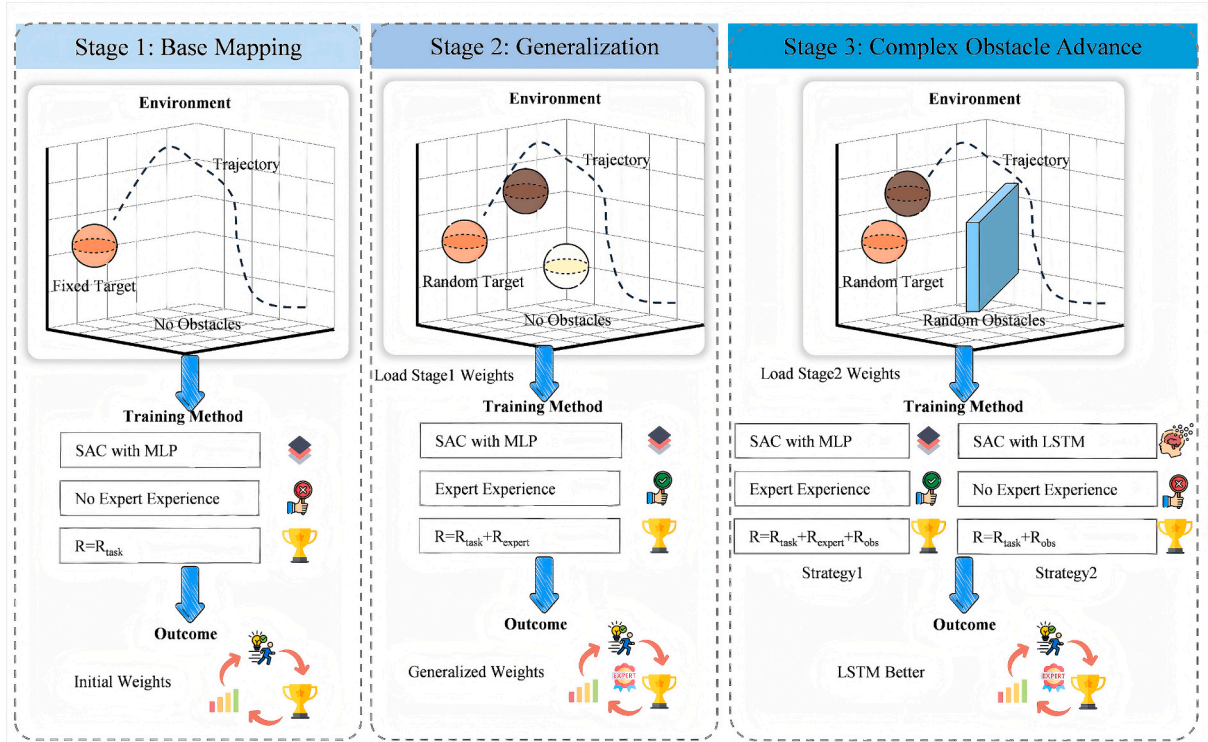


Fig. 2. three-stage course learning strategy.

network architecture to LSTM, compelling the agent to rely solely on internal memory and direct interaction with the environment to explore obstacle avoidance strategies, thereby validating the effectiveness of different mechanisms in complex scenarios.

2.3. Dynamics-validated dual experience replay mechanism

This paper proposes a hierarchical fusion strategy that decouples motion planning into two subproblems: trajectory generation and motion control. Synergizing the efficiency of traditional planning with the adaptability of reinforcement learning, the framework utilizes a simplified rigid-link model in the upper layer to generate collision-free expert demonstrations, while the lower layer acquires robust control policies. Specifically, Artificial Potential Fields (APF) are employed to synthesize high-quality reference trajectories, thereby guiding initial exploration and significantly enhancing sample efficiency.

2.3.1. Expert buffer construction via dynamics-validated trajectory

Three classical path planning algorithms, specifically the Artificial Potential Field (APF), Rapidly-exploring Random Tree (RRT), and Genetic Algorithm (GA), are employed to generate a substantial volume of initial trajectories based on a simplified rigid-body kinematic model. Specifically, APF utilizes virtual potential fields to guide the end-effector toward the target, RRT constructs a global search tree within the joint space via random sampling to address complex obstacle avoidance, and GA iteratively evolves globally optimal paths through population evolution. However, as the simplified model neglects cable tension coupling and deformation, these initial paths serve merely as candidate trajectories and do not strictly satisfy the physical constraints of the real system. Consequently, a dynamics screening mechanism is introduced

for preprocessing, where all candidate trajectories are subjected to forward dynamics resolution within the MuJoCo physics engine. Only trajectories that satisfy physical constraints are deemed valid samples and stored in the Expert Experience Replay Buffer.

The validated expert buffer operates in parallel with the Main Experience Replay Buffer, which stores data derived from the autonomous exploration of the agent, collectively constituting a dual experience replay architecture as illustrated in Fig. 3. During the training process, relying solely on a single main experience buffer imposes an exorbitant exploration cost on the agent within a high-dimensional space characterized by sparse rewards, often resulting in slow convergence or failure to converge. In contrast, the dual buffer architecture facilitates mixed sampling and a reward shaping mechanism. This mechanism computes the minimum Euclidean distance between the current state and expert trajectories to generate guidance rewards, thereby accelerating policy learning while ensuring physical feasibility.

2.3.2. Hierarchical computational strategy

MuJoCo provides high-fidelity physical simulation capabilities, particularly in the modelling of tendon dynamics. In MuJoCo, tendons are modelled using the site-to-site tendon primitive, which effectively simulates force transmission along the tendon and partially reproduces physical phenomena such as tendon vibration and elastic elongation. This capability is crucial for learning control policies that require consideration of temporal dependencies.

It is noteworthy that the highly underactuated nature and strong nonlinear coupling of cable-driven systems render direct inverse dynamics solving within the MuJoCo environment extremely challenging. First, accurate inverse dynamics resolution involves complex Jacobian inversion, where the computational time for a single solution is

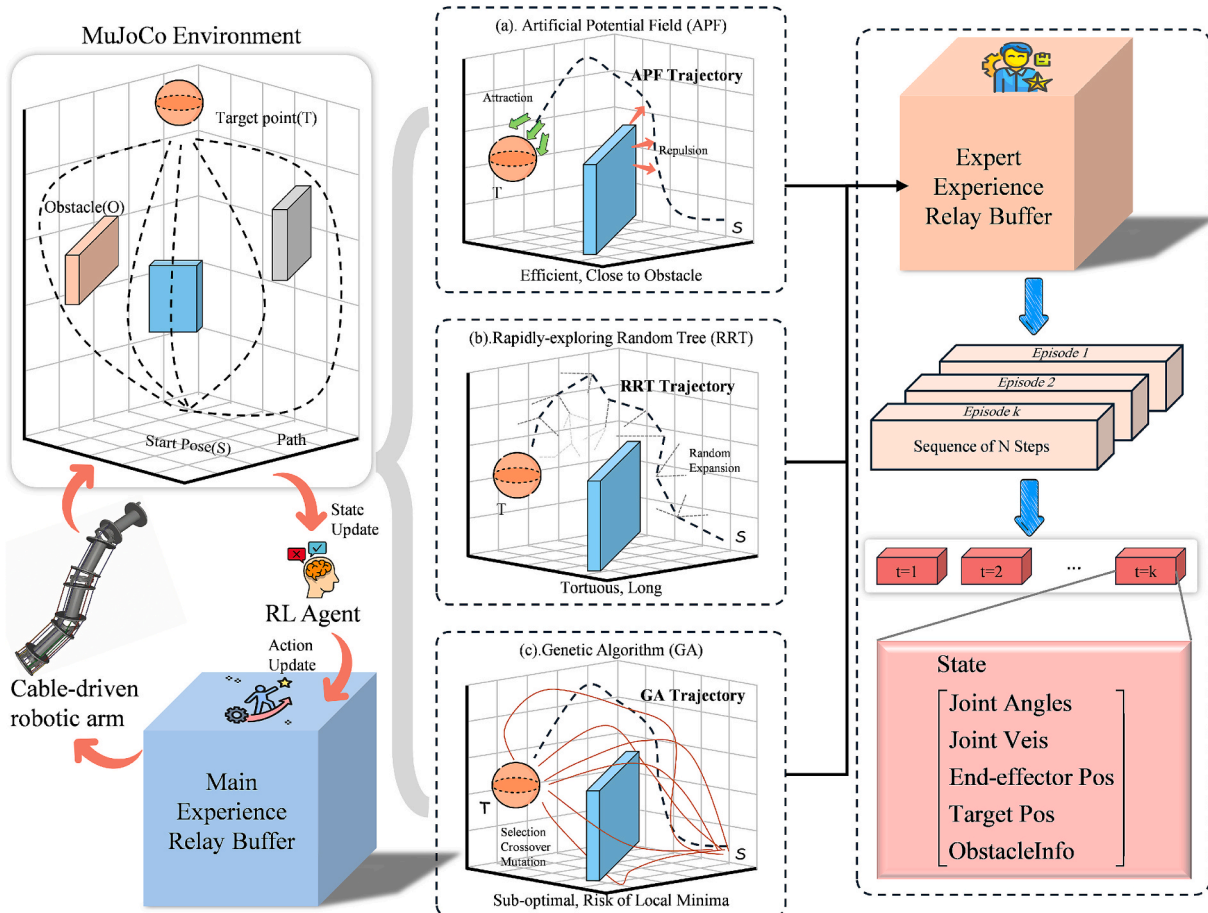


Fig. 3. Dual experience replay mechanism.

prohibitively high and fails to meet the real-time requirements for state transitions in reinforcement learning. Conversely, attempting pure reinforcement learning exploration without prior knowledge often results in the inability of the agent to effectively explore the valid state space. Therefore, the hierarchical strategy adopted in this paper effectively circumvents these issues. It leverages the simplified model, which has minimal computational cost, to rapidly generate a massive volume of candidate solutions, and subsequently employs MuJoCo for offline dynamic verification. The computational overhead of online inverse dynamics is circumvented while ensuring the high-fidelity physical executability of buffered samples, thereby achieving an optimal balance between efficiency and control precision.

2.4. Long short-term memory neural network

To address the temporal dependency and partial observability inherent in cable-driven manipulator motion planning, this paper introduces a Long Short-Term Memory (LSTM) network into the standard SAC algorithm, constructing an improved reinforcement learning architecture with explicit temporal modeling capability. The limitations of conventional MLP-based SAC in capturing historical dependencies are effectively addressed, significantly enhancing policy robustness and planning performance in complex dynamic environments.

In the motion planning process of cable-driven manipulators, the system's current state depends not only on instantaneous observations but also on the history of past states and action sequences. For instance, factors such as cable-induced vibrations and the manipulator's own inertia introduce time-dependent dynamics. The standard Soft Actor-Critic (SAC) algorithm employs a multilayer perceptron (MLP) as its core network architecture; however, when applied to such partially observable Markov decision processes (POMDPs), it often converges to suboptimal policies due to its inability to retain and utilize historical information, causing the planning process to stall.

To address this limitation, a Long Short-Term Memory (LSTM) network is integrated into the SAC framework, as illustrated in Fig. 4. Specifically, the current observation vector is first normalized via a layer normalization module to ensure consistent scale across dimensions. It is then passed through two MLP layers to project it into a lower-

dimensional latent space suitable for LSTM processing. A multi-layer unidirectional LSTM is employed—chosen because decision-making can only rely on past and present states, not future information. The resulting hidden state from the LSTM is subsequently fed into a fully connected network to learn nonlinear combinations of the temporal features, producing a compact feature vector. This feature vector is then passed to the subsequent layers of the SAC algorithm to compute the final control action and estimate state values.

In the architecture proposed in this study, the Long Short-Term Memory (LSTM) module serves as a temporal feature encoder bridging the partially observable environment and the Soft Actor-Critic (SAC) decision policy. By processing a sequence of 30 raw observation frames, the LSTM implicitly recovers unobservable high-order dynamic features from the position sequence, such as instantaneous velocity, acceleration, and the inherent inertia of the manipulator. These features are typically absent in single-frame static observations. Furthermore, its recurrent memory mechanism effectively captures the hysteresis effects and actuation delays inherent in cable transmission by retaining contextual information regarding past actions. Ultimately, by outputting a state vector containing comprehensive observational information, the LSTM transforms the original Partially Observable Markov Decision Process (POMDP) into a tractable Markov Decision Process (MDP). This transformation enables the Actor and Critic networks to derive optimal control decisions.

Regarding network architecture and parameter configuration, a stacked two-layer LSTM structure followed by a multi-head self-attention mechanism was selected. The two-layer configuration ensures effective extraction of high-level abstract temporal features while mitigating potential vanishing gradient problems associated with deeper networks. Consequently, convergence stability is effectively maintained amidst the non-stationary training signals inherent to reinforcement learning.

The incorporation of the attention mechanism enhances the capacity of the network to capture long-range state dependencies. This allows the network to attend directly to critical state transitions across time steps within the historical sequence and effectively alleviates the information decay typically observed in traditional LSTMs over long sequences. Additionally, the input sequence length is set to 30 frames to ensure

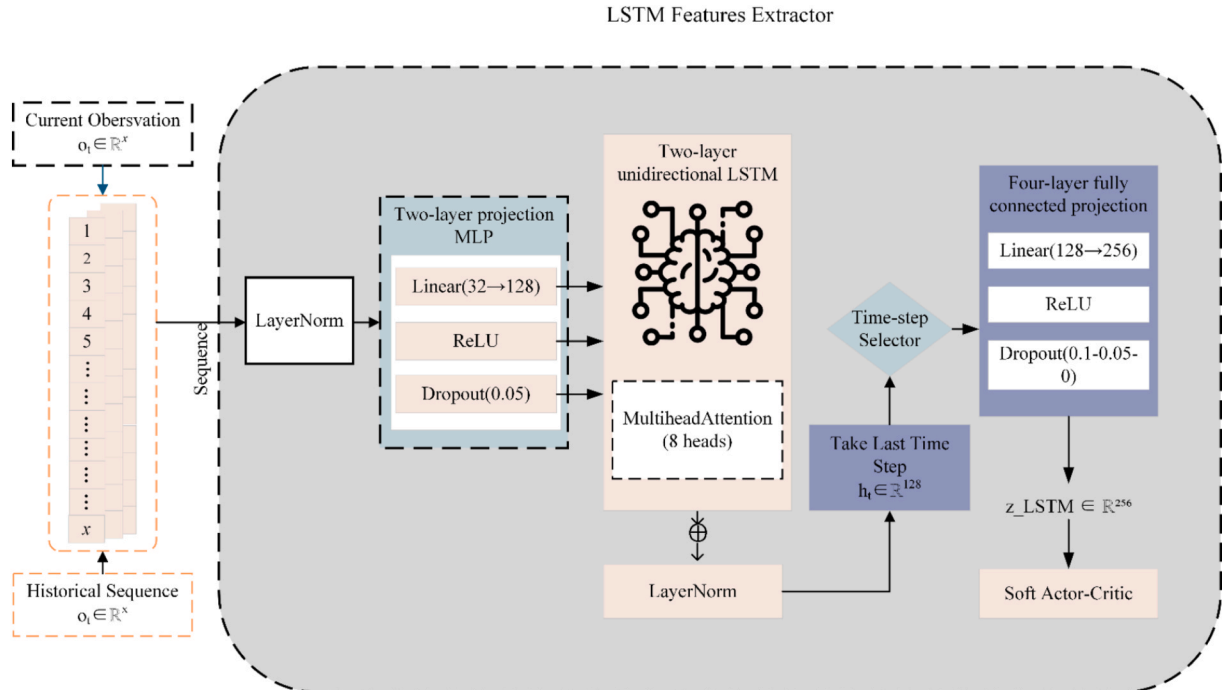


Fig. 4. LSTM features extractor.

coverage of the dynamic characteristics within the actuation system of the cable-driven manipulator. An optimal balance is achieved between the effective recovery of partially observable state information and the maintenance of computational efficiency.

2.5. Reinforcement learning control method based on SAC

The SAC algorithm is an off-policy method based on the maximum entropy reinforcement learning framework. This paper introduces an improved Soft Actor-Critic algorithm that integrates expert-guided experience. Built upon the standard SAC framework, the proposed method effectively accelerates policy convergence and improves task success rates through reward shaping and a dual experience replay mechanism.

This paper formulates the manipulator path planning and control problem as a Markov Decision Process (MDP), which consists of five core components:

$$(S, A, P, R, \gamma) \quad (1)$$

Where S denotes the state space, which provides the agent with partial information regarding its own pose, motion trend, and environmental layout. Where A denotes the action space, where each action corresponds to the control signals for the n drive motors.

Where P denotes the state transition probability, representing the probability of transitioning from state S to state $\hat{S}$ upon executing action. The discount factor γ is used to balance immediate rewards against future rewards by exponentially decaying the value of future returns. This attenuation ensures stable and efficient training of the algorithm.

The reward function R assigns an immediate scalar reward r to the agent for completing the state transition.

The specific formulation of the reward function is as follows:

$$r_{dist} = -2.0 \bullet d \quad (2)$$

$$r_{ctrl} = -0.001 \sum_j \tilde{a}_j^2 \quad (3)$$

$$r_{succ} = \begin{cases} 150 & d < threshold \\ 0 & otherwise \end{cases} \quad (4)$$

$$r_{path} = \begin{cases} 8.0 & The path is blocked \\ 0 & otherwise \end{cases} \quad (5)$$

r_{dist} (Eq.(2)) functions as a base distance penalty, utilizing the Euclidean distance d measured between the current end-effector position and the target. Employing linear negative feedback based on this metric ensures a constant gradient toward the goal, effectively mitigating the gradient vanishing problem that frequently hinders long-distance planning.

r_{ctrl} (Eq.(3)) serves as a control penalty targeting the acceleration $\tilde{a}_j$ of the j joint, given that excessive or rapidly fluctuating control inputs are prone to inducing high-frequency tendon oscillations and pronounced end-effector jitter. Although preliminary experiments with a coefficient of -0.01 resulted in an overly conservative agent with restricted action amplitudes, the subsequent adoption of a reduced magnitude successfully balances motion smoothness with task feasibility.

r_{succ} (Eq.(4)) denotes the success reward, where the criterion for task completion is defined by a Euclidean distance threshold of 0.04 m between the end-effector and the target. This value is determined not merely by positioning accuracy but also by the physical length of the end-effector, thereby reserving sufficient geometric margin to ensure reliable physical contact during real-world operation.

r_{path} (Eq.(5)) introduces a path-aware reward framework designed to penalize obstructed planning paths while concurrently promoting active navigation. In contrast to conventional approaches that limit penalties to instances of direct collision, a strategy which frequently entraps the policy within local optima near obstacle boundaries, the proposed

mechanism grants a positive reward when effective detouring maneuvers are generated following the detection of a blocked straight-line path. By actively encouraging agent to explore feasible bypassing trajectories, this design fosters proactive decision-making rather than relying on passive reactions to environmental constraints.

To accelerate policy learning and guide the search toward optimal solutions by leveraging expert demonstrations, the standard Soft Actor-Critic (SAC) algorithm is enhanced through integration of expert knowledge into the SAC Bellman update equation via reward shaping and the introduction of a dual experience replay buffer architecture.

At the onset of training, expert trajectories—comprising states s_{expert} actions, rewards, and subsequent states—are preloaded into a dedicated expert experience buffer. Concurrently, online interaction data $(s_t, a_t, r_t, s_{t+1}, d_t)$ collected through the agent's exploration of the environment, are stored in the primary experience replay buffer.

Since the state space comprises physical quantities of different natures, such as joint angles, motor velocities, and end-effector positions, directly inputting these raw values into the neural network would cause dimensions with larger numerical magnitudes to dominate the gradient propagation direction. During the preliminary exploration phase of this study, training was attempted using raw physical quantities as inputs. The results demonstrated that due to the significantly larger numerical scale of motor velocities compared to position changes, the policy network failed to capture subtle position changes, leading to training divergence.

All physical quantities in the s_t are normalized, with their specific definitions as follows:

$$s_t = [qpos_t, qvel_t, motor_{speed_t}, tendon_{length_t}, ee_t, tgt_t, obstacle_{coords_t}] \in \mathbb{R}^x \quad (6)$$

At each time step t , a state is constructed to encapsulate the manipulator's comprehensive proprioceptive dynamics and the surrounding environmental context, incorporating the kinematic configuration defined by joint angles $qpos_t$ and angular velocities $qvel_t$ of the i manipulator joints, Motor speeds $motor_{speed_t}$ of the n drive motors, Tendon lengths $tendon_{length_t}$ of the n tendons. Furthermore, to facilitate precise task execution and obstacle avoidance, the representation integrates the 3D Cartesian coordinates of the end-effector ee_t , 3D Cartesian coordinate tgt_t of the task target point, 3D Cartesian coordinate $obstacle_{coords_t}$ of the m obstacles in the scene, resulting in a total state dimensionality of $x = 2i + 2n + 3m$.

At each time step t , an action a_t is an n -dimensional continuous vector:

$$a_t = [v_1, v_2, \dots, v_n]_t \in \mathbb{R}^n \quad (7)$$

During each gradient update step, the algorithm uniformly samples mini-batches of transitions from both buffers. These sampled transitions are then processed through a reward shaping function, formally defined as follows:

$$R_{shape}(s_i) = \exp(-k \bullet d_{min}(s_i)) \quad (8)$$

where $d_{min}(s_i)$ denotes the minimum Euclidean distance between the state s_i and the expert states $s_{expert} \in S_{expert}$ in the expert experience buffer. The specific formula for $d_{min}(s_i)$ is as follows:

$$d_{min}(s_i) = \min_{s_{expert} \in S_{expert}} \|s_i - s_{expert}\|_2 \quad (9)$$

By linearly combining the environment-provided reward r_t with the shaping reward $R_{shape}(s_i)$ a composite reward signal r_t is formed:

$$r_t = r_t + \omega_{shape} \bullet R_{shape}(s_i) \quad (10)$$

where ω_{shape} is a hyperparameter that modulates the strength of guidance from expert demonstrations.

The composite reward r_t is subsequently used to construct the target Q-value y_t :

$$y_t = r_t + \gamma(1 - d_t)(\min_{i=1,2} Q_{\theta_i}(s_{t+1}, a_{t+1} - \alpha \log \pi_{\phi}(a_{t+1}|s_{t+1})) \quad (11)$$

Subsequently, the critic loss is minimized, with the loss functions for the two Q-value networks remaining as the mean squared Bellman error:

$$L_{Q_i}(\theta_i) = E \left[(Q_{\theta_i}(s_t, a_t) - y_t')^2 \right] \text{ for } i = 1, 2 \quad (12)$$

By minimizing the gradient loss, the Q-value networks learn to approximate the environmental rewards and guide the agent toward emulating expert demonstrations. The update rule for the policy network remains unchanged:

$$L_{\pi}(\phi) = E_{S_t, D} \left[\alpha \log \pi_{\phi}(a_t|s_t) - \min_{i=1,2} Q_{\theta_i}(s_t, a_t) \right] \quad (13)$$

Since the learning objective of the Q-value function has already been shaped by expert demonstrations, the policy network, during its update process, produces strategies that both imitate expert demonstrations and accomplish the environmental task in order to generate actions with higher Q-values. Furthermore, the temperature parameter and the target network updates remain consistent with those of the standard SAC algorithm.

3. Case study

To systematically evaluate the effectiveness of the proposed curriculum learning framework for complex obstacle-avoidance tasks with a cable-driven redundant manipulator, this section details the methodology employed, the specific training parameters of the reinforcement learning agent, and the configurations of each stage of the curriculum learning process.

3.1. Model design

The cable-driven manipulator investigated in this paper is a high-degree-of-freedom serial flexible manipulator. Its initial three-

dimensional model was parametrically constructed in SolidWorks to provide a foundation for subsequent dynamic simulations. The core architecture consists of four rigid links sequentially connected by universal joints, with a total arm length of 598.5 mm. As shown in Fig. 5(a), the manipulator is actuated cooperatively by 12 drive tendons, forming an underactuated system. These 12 tendons originate from the base platform and are routed through 12 uniformly distributed guide holes on the end disc of each link, where each link has a length of 153 mm. The configuration and end-effector position of the manipulator are controlled by adjusting the lengths of these 12 drive tendons. The materials and physical parameters of the components are listed in Table 2 and Table 3.

3.2. Model conversion

To construct a simulation environment capable of accurately reflecting the complex dynamic characteristics of the cable-driven manipulator, the parameterized SolidWorks model is converted into MuJoCo's MJCF format. Furthermore, as shown in Fig. 5(b), the mass and inertia properties of the links and cables, derived from the material parameters listed in Table 2 and Table 3, are also incorporated into the MuJoCo model to ensure the fidelity of the dynamic simulation.

3.3. Expert experience parameter configuration

To comprehensively evaluate the robustness of the proposed

Table 2
Materials for robotic arm components.

Part	Material
Universal joint	Aluminum alloy
Steel wire rope	Stainless steel
Platform	Aluminum alloy
Rigid joint	Aluminum alloy

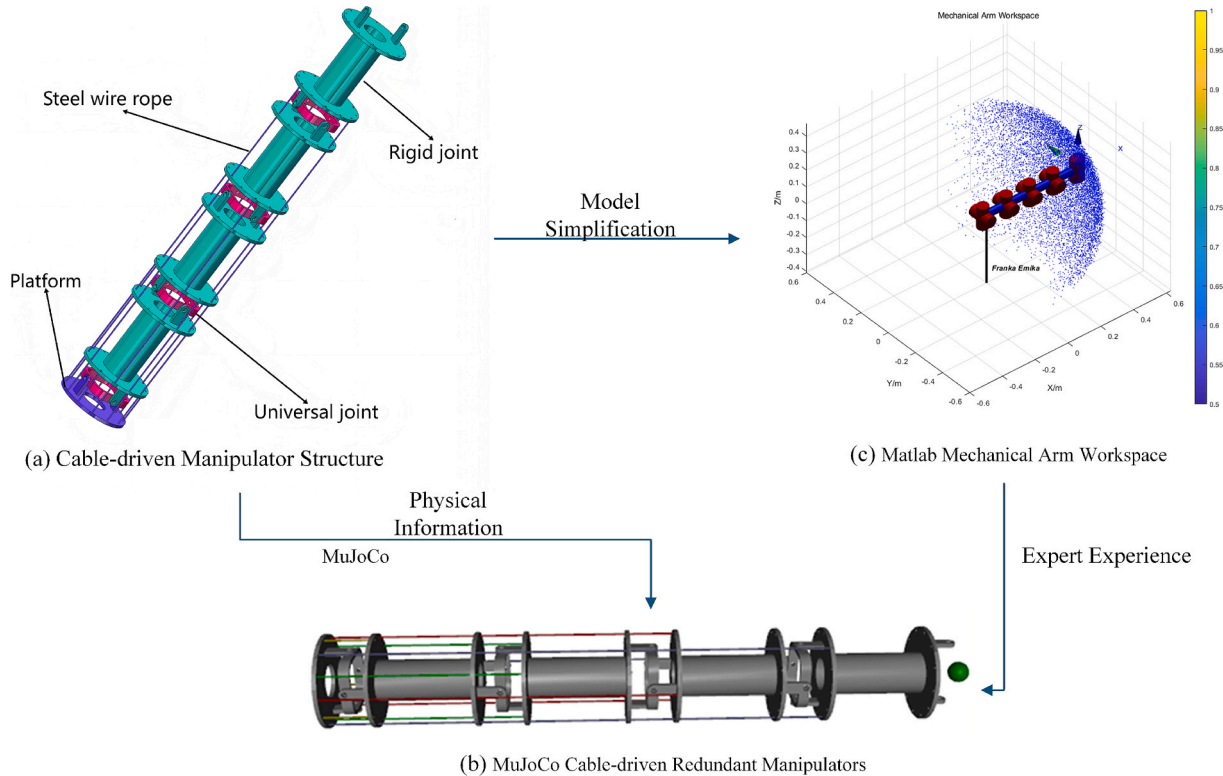


Fig. 5. Structural design, workspace, and high-fidelity simulation model of the cable-driven manipulator.

Table 3
Material parameters.

Material	Density ($\rho/(kg/m^3)$)	Poisson's Ratio (μ)	Elastic Modulus (E/GPa)
Aluminum alloy	2750	0.33	70
Stainless steel	8000	0.28	193

framework, three path planning algorithms with distinct characteristics are employed to generate expert demonstrations.

The APF method is a local planning approach that guides manipulator motion by constructing virtual forces. Rapidly-exploring Random Tree (RRT) is a sampling-based global planning algorithm that incrementally builds a search tree by randomly expanding in the state space to find a feasible path. The Genetic Algorithm (GA) is a bio-inspired global optimization technique that iteratively improves candidate paths through selection, crossover, and mutation operations applied to a population of path waypoints. All three algorithms are executed on the same kinematic model, with their key parameter configurations listed in Table 4, Table 5, and Table 6.

The expert demonstration generation described in this section is conducted on a simplified four-link, eight-joint kinematic model constructed using Denavit–Hartenberg parameters, as illustrated in Fig. 5 (c). The structure comprises four serial links, with the motion range of each of the eight joints constrained to ± 0.5235 rad. Target points for the end-effector are randomly sampled within a predefined three-dimensional workspace bounded by [0.52, 0.12, 0.05] and [0.55, 0.15, 0.08]. Obstacles are modelled as spheres, with their positions and radii consistent with those in the training environment.

3.4. Parameter configuration of reinforcement learning agent

All experiments in this paper were conducted on a dedicated computing workstation with the following specifications: an Intel i5-12600KF central processing unit and an NVIDIA RTX 4070 Super graphics processing unit. A complete three-stage curriculum learning run required approximately four hours of total training time.

As shown in Table 7, the optimal LSTM configuration balancing efficacy and computational overhead was determined through sensitivity analysis; Table 8 presents the RL parameters, based on literature benchmarks and fine-tuned for high-dimensional characteristics to balance stability and sample efficiency.

3.5. Ablation experiment

To comprehensively validate the proposed algorithm, this section presents the experimental setup for multi-level ablation studies. To clearly distinguish between model structures under different experimental configurations, a unified naming convention is defined as shown in Fig. 6 and Table 9. The nomenclature represents the expert experience ratio or network architecture adopted in the second and third training stages, respectively. The specific training parameter settings for each stage are detailed in Table 10.

Based on the established nomenclature, an ablation analysis focusing on the strategy of introducing expert demonstrations was conducted first. To verify the compatibility of the algorithm with different prior

Table 4
APF parameter configuration.

Parameter symbols	Physical meaning	Numerical value
ζ	Gravitational gain coefficient	0.3
η	Repulsive gain coefficient	0.2
ρ_0	Repulsive influence range(m)	0.05
ε	Target arrival threshold(m)	0.015
N_{max}	Maximum number of iterations	1500

Table 5
Genetic algorithm parameter setting.

Parameter	Numerical value
Population size	15
Max generations	20
Mutation rate	0.1
Crossover rate	0.6
Num waypoints	6
Elite ratio	0.2
Tournament size	3

Table 6
RRT parameter setting.

Parameter	Numerical value
Max nodes	2000
Step size	0.1
Joint step	0.005
Max iter	1500

Table 7
LSTM parameter settings.

Parameter	Numerical value
Input dimension	52
LSTM hidden dimension	128
Number of LSTM layers	2
One-way LSTM	Ture
Attention mechanism	Ture
Sequence length	30

Table 8
Hyperparameter Settings.

Hyperparameter	Value	Reference
Learning rate	0.0001	A stability-focused setting for precise convergence. It aligns with the 10^{-4} magnitude used in (Ji, Gao, Xiao, & Sun, 2025) (0.00025), being slightly more conservative than (Su, Yuan, Qu, Wang, & Zhou, 2026) (0.0003) to prevent oscillations during long-horizon training.
Buffer size	700,000	Comparable to 1,000,000 used in (Su et al., 2026) and (Zhang et al., 2025).
Batch size	512	Consistent with Yang Bo et al. (Yang et al., 2023). Larger batches stabilize gradients compared to 256 used in Mao J et al. (Qi et al., 2024).
gamma	0.99	Standard setting for long-horizon tasks. Consistent with (Garg, Goharimanesh, Sajjadi, & Janabi-Sharifi, 2025), (Ji et al., 2025), and (Su et al., 2026).
tau	0.001	Ensures stable target updates. Similar magnitude to 0.005 used in Yang Bo et al. (Li et al., 2024) and Su M et al. (Su et al., 2026)
Ent coef	auto	Automatic entropy tuning is preferred for robustness over fixed values, as validated by (Li et al., 2024), (Zhang et al., 2025), and (Qi et al., 2024).
Train freq	1	Standard per-step update frequency, consistent with (Yang et al., 2023)
Gradient steps	1	Single gradient step per update iteration, consistent with (Yang et al., 2023)
Net arch	[256,256,256]	MLP architecture matches the specific configuration of (Su et al., 2026).
Activation fn	SILU	This aligns with the advanced usage of SeLU in (Zhang et al., 2025), offering an improvement over the standard ReLU used in (Garg et al., 2025) while maintaining efficiency.

knowledge, traditional path planning algorithms based on varying

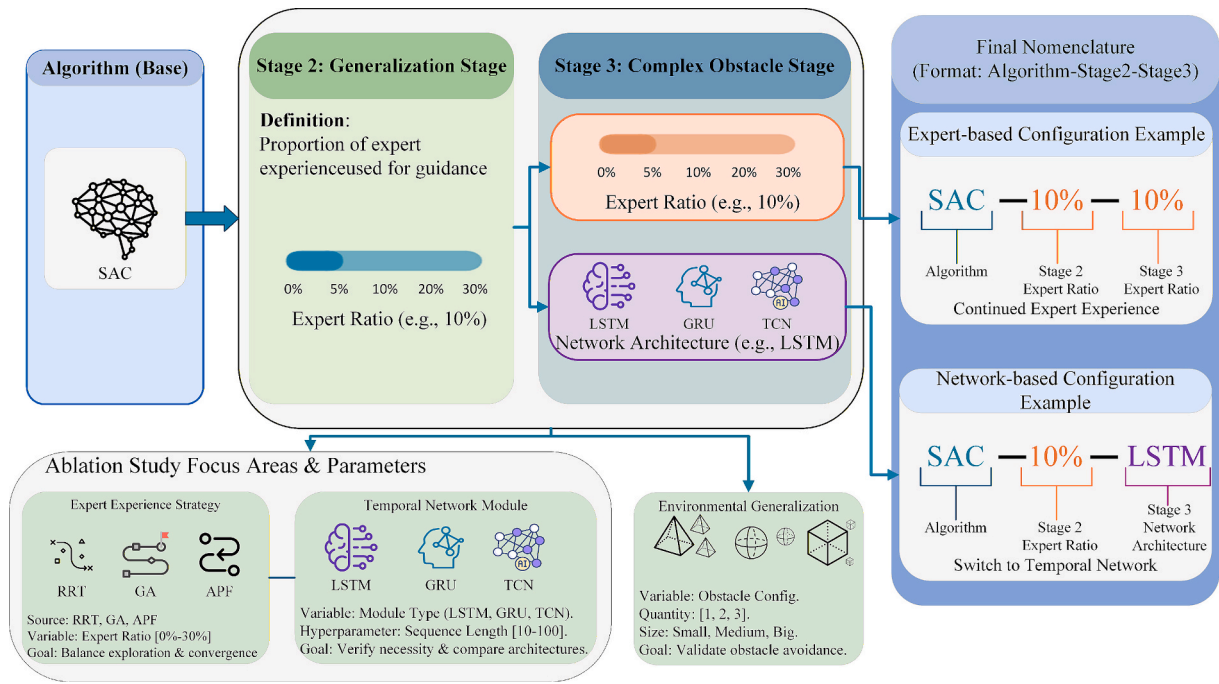


Fig. 6. Nomenclature Rules.

Table 9
Configuration of Ablation Study.

Category	Variable Name	Range
Expert Experience Strategy	Expert Ratio	[0%,5%, 10%, 20%, 30%]
Temporal Network	Module Type	LSTM, GRU, TCN
Network Hyperparameters	Sequence Length	[10, 20, 30, ..., 100]

Table 10
Course Learning Parameter Setting.

Configuration Item	Stage 1	Stage 2	Stage 3-Strategy 1	Stage 3-Strategy 2
TimeSteps	300,000	300,000	200000–300000	200000–300000
Target Type	Fixed	Random	Random	Random
Network	MLP	MLP	MLP	LSTM
Architecture				
Expert ratio	/	0%–30%	0%–30%	/

principles, such as sampling-based RRT, the evolutionary Genetic Algorithm, and the Artificial Potential Field method, were selected as sources for expert demonstrations. Furthermore, compatibility experiments introducing different ratios of expert demonstrations under the identical base algorithm were compared to determine the optimal expert auxiliary ratio that balances exploration efficiency with convergence speed.

Regarding the core temporal feature processing module, an in-depth component ablation was performed. On one hand, the necessity of the LSTM structure in feature processing was verified by testing the LSTM component under configurations with and without the perception function. On the other hand, the performance differences among various temporal network architectures, including LSTM, GRU, and TCN, in processing continuous state spaces were compared in detail. Additionally, a sensitivity analysis regarding the input sequence length of the LSTM was conducted to determine the optimal length that captures effective temporal dependencies while ensuring real-time inference.

To evaluate the generalization capability within complex environments, experimental scenarios targeting distinct obstacle attributes are

configured, specifically testing obstacle avoidance performance amidst varying sizes and quantities ranging from 1 to 3. The resulting multi-dimensional analysis substantiates the robustness of the proposed architecture and its high feasibility in addressing intricate path planning tasks.

4. Results and discussion

This section presents and analyzes the performance of the proposed curriculum learning framework, discusses the results of the ablation studies, identifies the factors influencing model training, and evaluates the effectiveness of the key components within the framework.

4.1. Performance evaluation of path planning algorithms for expert experience

This section evaluates the performance of three distinct classical path planning algorithms when employed as sources of expert knowledge to guide the reinforcement learning agent, through comparative experiments. A quantitative analysis is conducted based on performance metrics including average reward, path efficiency (measured by average episode length in steps), and path quality (assessed by average final and minimum distances to the target). This analysis investigates the intrinsic advantages and limitations of each guidance strategy. Moreover, all models are trained and evaluated under identical environmental conditions and with the same reward function to ensure a fair comparison. The experimental results are presented in Table 11.

The RRT-guided SAC achieves the highest average reward (396). As a probabilistically complete algorithm, RRT is capable of finding a feasible path in complex environments where a solution exists. This enables the

Table 11
Experimental results.

Metric	RRT-guided SAC	GA-guided SAC	APF-guided SAC
Average Reward	396	171	95
Average Steps	335	159	66
Average Final Dist.	0.038 m	0.032 m	0.029 m
Average Min. Dist	0.03 m	0.03 m	0.029 m

agent to effectively avoid substantial negative rewards caused by failed exploration or entrapment in local optima, thereby maximizing the cumulative reward. However, this model exhibits a significantly higher average episode length of 335 steps, substantially exceeding those of the other models. The fundamental reason lies in the fact that RRT prioritizes rapid feasibility over optimality; consequently, the expert trajectories it generates are often highly stochastic and suboptimal, exhibiting excessive length, tortuosity, and lack of smoothness. When the agent imitates such low-quality demonstrations, its policy inherits this inefficiency, leading to a sharp increase in the number of decision steps. This highlights the pronounced deficiency of RRT in terms of path efficiency and execution cost.

Compared with RRT, the Genetic Algorithm (GA) iteratively evolves higher-fitness path solutions through selection, crossover, and mutation operations within a population of candidate paths. Consequently, decision efficiency is significantly improved, with the average episode length (159 steps) reduced by 52% relative to RRT, demonstrating the direct impact of high-quality guidance data on enhancing decision efficiency. However, as a heuristic optimization method, GA does not guarantee global convergence. In environments featuring complex obstacle configurations, as illustrated in Fig. 7, the GA population may converge prematurely to a local optimum or even fail to discover any feasible path. This uncertainty in convergence leads to a higher task failure rate compared to RRT, thereby reducing the average reward (171.321). Furthermore, GA typically incurs higher computational costs due to the need to maintain and evolve an entire population, resulting in a slower expert demonstration generation speed (2.5 h) compared to the other two methods.

The Artificial Potential Field (APF) method provides the SAC agent with a highly smooth, near-shortest, idealized trajectory. This form of guidance instills in the agent an exceptionally efficient and direct behavioral pattern, enabling the model to complete the task in only 66 steps—substantially fewer than GA (159 steps) and RRT (335 steps). Moreover, the APF-guided model achieves the best performance in terms of both average final distance and minimum distance to the target (both 0.029 m). To achieve minimal path length, APF-based planning tends to steer the trajectory closely along the safety margins of obstacles. While the presence of the repulsive field theoretically guarantees collision avoidance, the planned waypoints remain near the critical threshold at which repulsive forces are activated. From the perspective of the reward function, this behavior is interpreted as a persistently high-risk state, causing the APF-guided SAC agent to accumulate continuous safety-related penalties throughout most of the trajectory. In contrast, the trajectories generated by RRT and GA adopt more conservative strategies that maintain greater clearance from obstacles, thereby avoiding this penalty component.

The Artificial Potential Field (APF)-guided reinforcement learning model is selected for subsequent training, characterized by the lowest average episode length and the smallest final distance to the target, attributes critical for applications demanding real-time responsiveness

and high precision. The relatively lower average reward of the APF-guided SAC model is not attributable to task failure, but rather to its path-shortening execution strategy, which incurs safety-related penalties; this can be mitigated through fine-tuning of the reward function and optimization of algorithmic parameters. Furthermore, as shown in Table 12, APF requires only 30 min to generate 3,000 expert trajectories, significantly faster than RRT (1 h) and GA (2.5 h).

4.2. Performance analysis of expert-based agent in small-obstacle environments

In Stage 3—Strategy 1 of the curriculum learning framework, the experimental environment is configured with two small-scale obstacles to investigate the effect of expert guidance. Notably, the obstacle-aware reward term is not employed during training in this stage.

The analysis demonstrates that expert guidance plays a decisive role in high-dimensional, complex obstacle-avoidance tasks. In the absence of such prior knowledge, the SAC-0%–0% agent—relying solely on autonomous exploration—fails to discover any viable obstacle-avoidance policy throughout Stage 3, due to the vast state-action space and sparse reward signals. Its evaluation average reward curve remains nearly flat in the mid-to-late training phase, persistently hovering in the negative region, indicating a complete stagnation of learning. Consequently, the final average distance reaches 0.270 m, far exceeding the success threshold of 0.04 m.

As shown in Fig. 8, all experimental groups incorporating expert demonstrations in Stage 3 achieve superior performance in both average reward and final distance compared to the pure RL baseline [Fig. 8(a)] confirming that the reward shaping mechanism effectively guides the agent. The relationship between performance and the proportion of expert experience is non-monotonic. In this complex obstacle-avoidance task, performance improves as the expert ratio increases from 0% to 25%, peaking at an average reward of 886. However, further increasing the ratio to 30% leads to performance degradation, with a sharp decline observed at 30% (average reward: −487). At this ratio, the model's final average distance rises to 0.055 m, offering only marginal improvement over the pure RL model.

In Stage 2, all models undergo a rapid decline in average episode length during the early training phase (0–100 k steps). From the mid to late phase (100 k–300 k steps), most groups exhibit varying degrees of recovery or oscillation, as illustrated in Fig. 9. Specifically, the SAC-20%–20% [Fig. 9(d)], SAC-5%–5% [Fig. 9(b)], and SAC-25%–25%

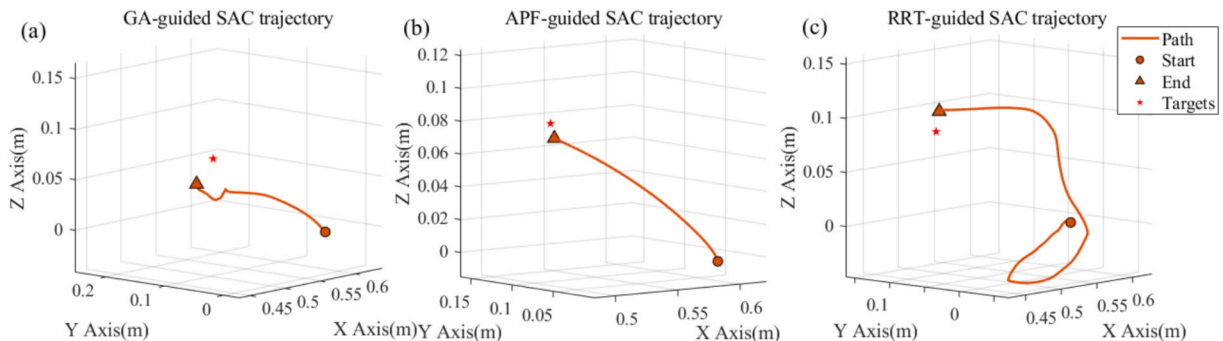


Fig. 7. APF\GA\RRT-guided SAC trajectory.

Table 12

Expert experience generation time.

Algorithm	Generation Time
APF	30 min
RRT	1 h
GA	2.5 h

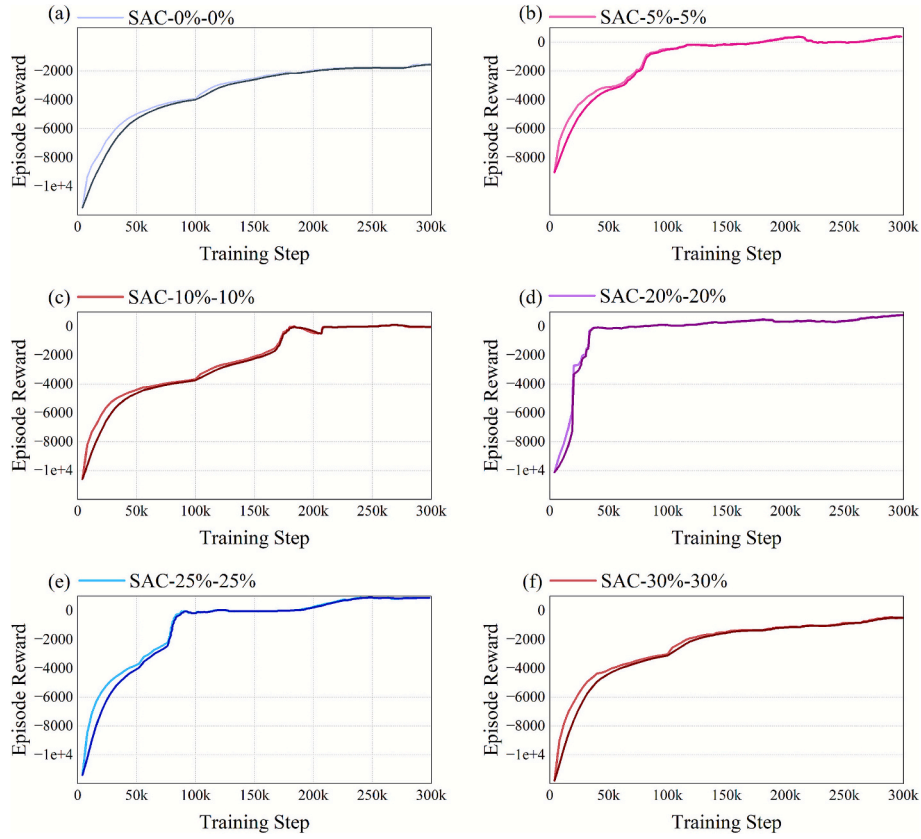


Fig. 8. Stage 3 model performance.

[Fig. 9(e)] groups, despite their sharp initial decline, all demonstrate noticeable recovery in the mid-to-late training period.

Notably, SAC-20%-20% and SAC-5%-5% converge to similar average episode lengths by the end of training, but for fundamentally different reasons: the former suffers from insufficient expert guidance, leading to uncontrolled exploration, whereas the latter experiences policy locking due to an excessively high proportion of expert data. The SAC-30%-30% model [Fig. 9(f)] exhibits an initial decrease followed by a subsequent increase in average episode length, which remains consistently high throughout training. This indicates that while a 30% expert ratio accelerates early convergence, it severely constrains the agent's capacity for autonomous exploration and policy refinement, ultimately resulting in strategy degradation during the mid-to-late stages.

It is worth highlighting that the SAC-10%-10% model [Fig. 9(c)] achieves the best performance in Stage 2, with its average episode length reduced to 620 steps—significantly lower than all other fixed-ratio groups. This suggests that a 10% expert ratio provides sufficient prior knowledge while preserving the agent's ability to continuously optimise its action sequences and discover efficient trajectories. In contrast, although the SAC-0%-0% model [Fig. 9(a)] initially displays the shortest average episode length, its curve exhibits severe fluctuations and a substantial rebound in the later phase, ultimately exceeding 600 steps. This behavior stems from the ineffectiveness of its policy, which fails to meaningfully reduce the distance to the target, causing the environment to terminate episodes prematurely and thereby yielding deceptively short average episode lengths.

As listed in Table 13, the hybrid expert model achieves the best performance in terms of average reaching distance, attaining a value of 0.029 m—outperforming all fixed-ratio groups (the best of which are the 5% and 25% models, both at 0.030 m). Compared to the no-expert model (0.270 m), the hybrid model demonstrates an accuracy improvement of approximately 89%, significantly enhancing the agent's

performance. Furthermore, it yields a performance gain of approximately 21.6% over the high-ratio SAC-30%-30% model (0.055 m). This indicates that the hybrid expert model not only inherits the exploration advantages of low-ratio models but also acquires the convergence stability of moderate-ratio models, thereby achieving near-global-optimal precision in complex tasks.

The aforementioned observations demonstrate that a low proportion of expert experience (5%–10%) provides essential guidance while allowing the agent sufficient freedom to explore the policy space, enabling it to refine and even surpass the expert policy. A moderate proportion (20%) strikes a balance between accelerating convergence and preserving exploratory capability. In contrast, a high proportion (30%) leads to excessive reliance on expert actions, which compresses the policy distribution, reduces entropy, and suppresses exploration of potentially superior trajectories, ultimately resulting in policy locking. Although the 30% model exhibits the fastest reward growth in the early phase of Stage II (reaching a final reward of 1856), its performance deteriorates sharply in Stage III, indicating that high-ratio guidance can accelerate learning in simpler tasks but compromises policy generalization in more complex scenarios.

Consequently, the dynamic strategy proposed in this study leads to the following conclusion: expert experience should be capped at 30% and dynamically adjusted according to the curriculum stage and task complexity—employing a lower ratio of 5%–10% in simpler stages to encourage exploration, and a higher ratio of 20%–30% in more complex stages to effectively guide learning. Ultimately, the hybrid-ratio model achieves an average reaching distance of 0.029 m, confirming its superiority in learning high-precision policies.

4.3. Effectiveness of LSTM module design in medium-obstacle environments

In Stage III—Strategy 2 of the curriculum learning framework, the

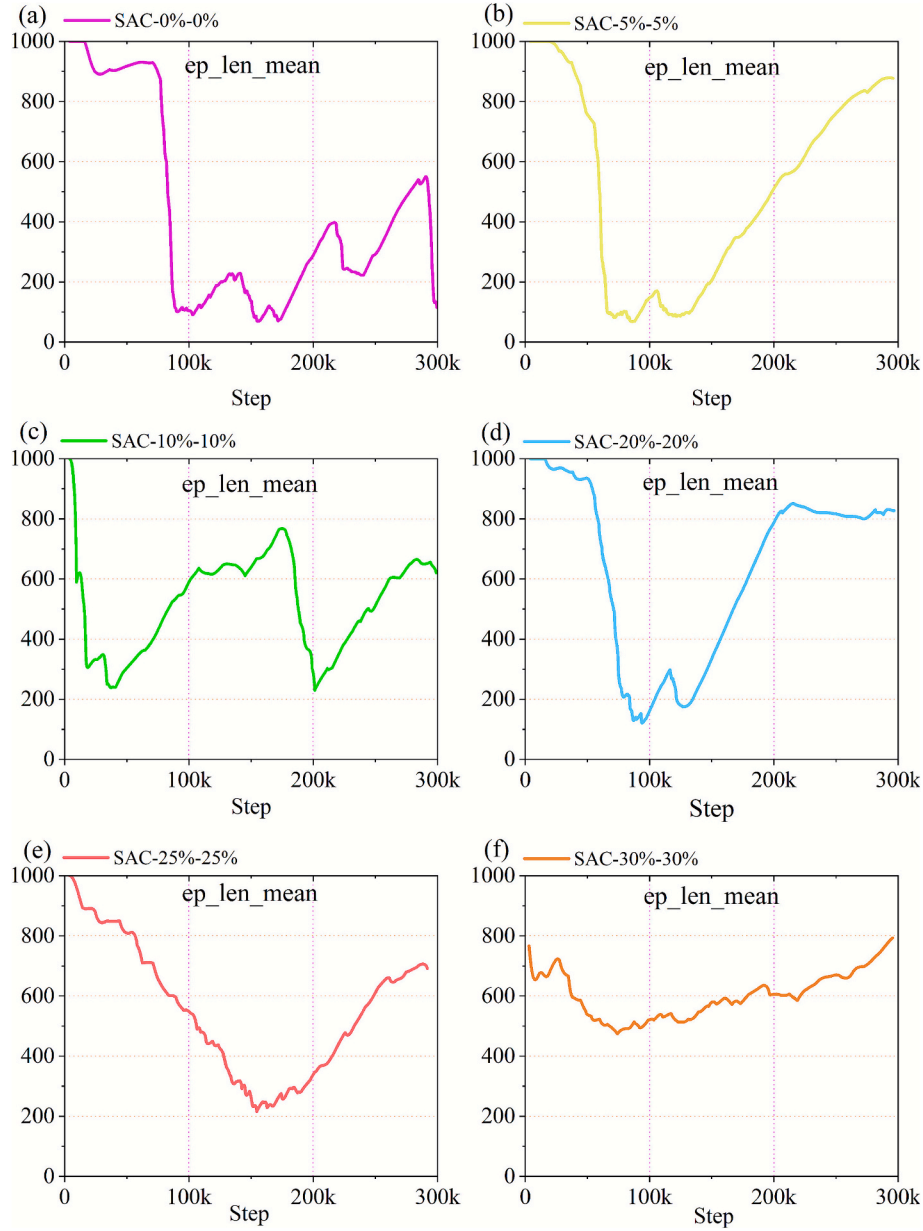


Fig. 9. Stage 2 model performance.

Table 13
SAC Agent Performance.

Agent	Episode Reward	Average Final Dist	Average Step Count
SAC-0%-0%	-1,529	0.270 m	1000
SAC-5%-5%	391	0.034 m	192
SAC-10%-10%	-33	0.054 m	493
SAC-20%-20%	789	0.036 m	129
SAC-25%-25%	886	0.038 m	252
SAC-30%-30%	487	0.055 m	593
SAC-10%-20%	95	0.029 m	66

experimental environment is configured with two large-scale obstacles, and the training in this stage incorporates the obstacle-aware reward function. The experimental results indicate that only two models—SAC-10%-LSTM [Fig. 10(d)] and SAC-10%-20% [Fig. 10(e)]—successfully complete the obstacle-avoidance task.

Among them, SAC-10%-LSTM refers to the agent that received policy guidance with 10% expert experience during Stage II and, in Stage III,

employed a Long Short-Term Memory (LSTM) network to enhance its capacity to retain and process sequences of historical states. This architecture effectively models the temporal dependencies inherent in unstructured, complex environments. In the obstacle-avoidance evaluation, SAC-10%-LSTM demonstrates superior performance, achieving an average final distance of 0.033 m, an average episode length of 886 steps, and an average reward of 33,400.

The SAC-10%-LSTM agent is capable of flexibly planning a smooth, safe, and efficient three-dimensional path. The LSTM module effectively encodes state-action pairs from multiple previous time steps, enabling the agent to recognize the boundary contour of obstacle Obs2 as it approaches. Consequently, it elevates its end-effector along the Z-axis and selects a globally optimal route that passes over both obstacles, accurately reaching the target without any collisions. This result confirms that the LSTM network enables the agent to capture and exploit long-range temporal dependencies in environmental states.

The optimal expert-ratio model, SAC-10%-20%, also succeeds in obstacle avoidance but exhibits significantly inferior performance compared to SAC-10%-LSTM, achieving an average final distance of

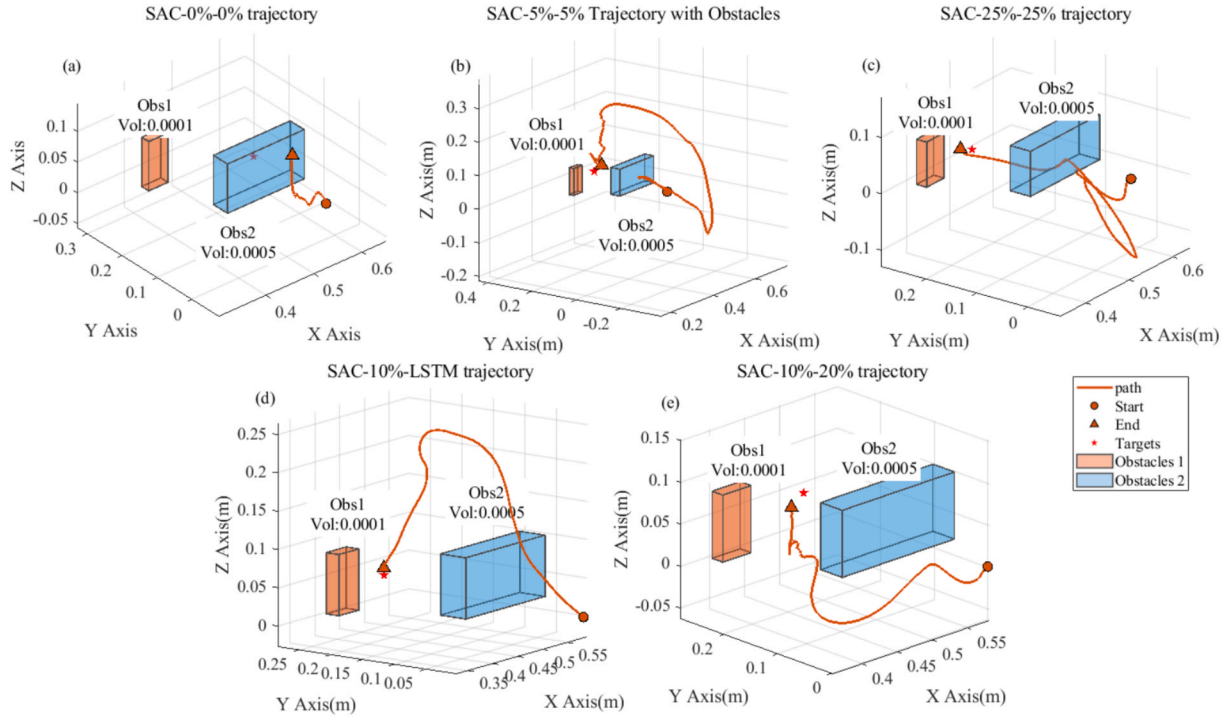


Fig. 10. Agent trajectory in medium-obstacle environments.

0.051 m, an average episode length of 700 steps, and an average reward of 22,000. Its planned trajectory displays higher tortuosity; specifically, the agent exhibits local oscillations when approaching the large obstacle Obs1 and performs pronounced tentative detours near Obs2. Unlike SAC-10%-LSTM, SAC-10%-20% does not incorporate an LSTM module, and its policy network relies solely on the current observation for decision-making. Consequently, when confronted with large obstacles, it is unable to plan multi-step trajectories as effectively as SAC-10%-LSTM and must instead rely on immediate feedback for local corrections, resulting in a more convoluted and less efficient path.

Further analysis of other configurations reveals that both SAC-5%-5% [Fig. 10(b)] and SAC-25%-25% fail to generate fully collision-free trajectories. The SAC-5%-5% model, with only 5% expert experience, remains heavily dependent on random exploration during early training. In the high-dimensional continuous action space, this leads to repeated trials of meaningless action combinations, yielding poor policy initialization. In complex environments, the agent is prone to becoming trapped in locally suboptimal solutions—evidenced by collisions with Obs1 before redirecting toward the target. Compared to SAC-10%-LSTM and SAC-10%-20%, this model lacks memory of historical states, rendering it incapable of recognizing whether it is currently in a hazardous region or leveraging past successful avoidance experiences to inform its current decisions.

Conversely, the SAC-25%-25% model [Fig. 10(c)] suffers from an excessively high expert ratio (25%), which causes the policy network to become confined to a narrow subspace early in training. The network prioritizes minimizing deviation from expert actions over exploring diverse behaviors in the action space. This premature convergence limits the agent's ability to generate sufficient action variability for navigating around complex obstacles in later stages, resulting in policy locking. Specifically, during the attempt to circumnavigate Obs2, the planned trajectory collides with the obstacle due to insufficient exploratory flexibility.

As shown in Fig. 10(a), the pure reinforcement learning baseline model completely fails in this stage: the agent is unable to recognize obstacles, oscillates repeatedly in front of the target, and becomes stuck near the obstacles, failing to accomplish the obstacle-avoidance task.

This further demonstrates that, in high-dimensional, sparse-reward environments with multiple obstacles, relying solely on the agent's autonomous exploration is insufficient to converge to a feasible solution, necessitating the incorporation of structured learning mechanisms and prior knowledge for effective guidance.

The experimental results demonstrate that the SAC-10%-LSTM model, which integrates the curriculum learning framework, expert-guided training, and LSTM-based temporal modelling, exhibits superior obstacle-avoidance performance and robustness in complex obstacle-rich environments compared to both the pure reinforcement learning baseline and other expert-guided variants. This design effectively balances the trade-offs between exploration and exploitation, as well as between short-term reactivity and long-term planning.

4.4. Effectiveness of LSTM module design in medium-obstacle environments without perceptual function

In Stage III—Strategy 2 of the curriculum learning framework, the experimental environment is configured with two large-scale obstacles, and the obstacle-aware reward function is removed during training in this stage. While all other models fail to complete the obstacle-avoidance task, the SAC-10%-LSTM model remains capable of successful navigation, achieving an average final distance of 0.052 m without any collisions. As shown in Fig. 11(a), the agent's trajectory successfully circumvents both Obs1 and Obs2 to reach the target point. This indicates that, even in the absence of explicit obstacle-aware reward guidance, the agent can infer hazardous regions in the current environment and make appropriate avoidance decisions by leveraging the spatiotemporal patterns of previously successful obstacle-avoidance trajectories stored in its memory. As shown in Fig. 11(b) and Fig. 11(c), both the pure reinforcement learning baseline model (SAC-0%-0%) and the optimal fixed expert-ratio model (SAC-10%-20%) fail under this condition. SAC-0%-0%, lacking both expert guidance and memory of past interactions, falls into an oscillatory loop near the target point in the absence of explicit reward feedback. SAC-10%-20%, while achieving strong performance when trained with the obstacle-aware reward function and expert demonstrations, fails upon removal of the perception-based reward: its

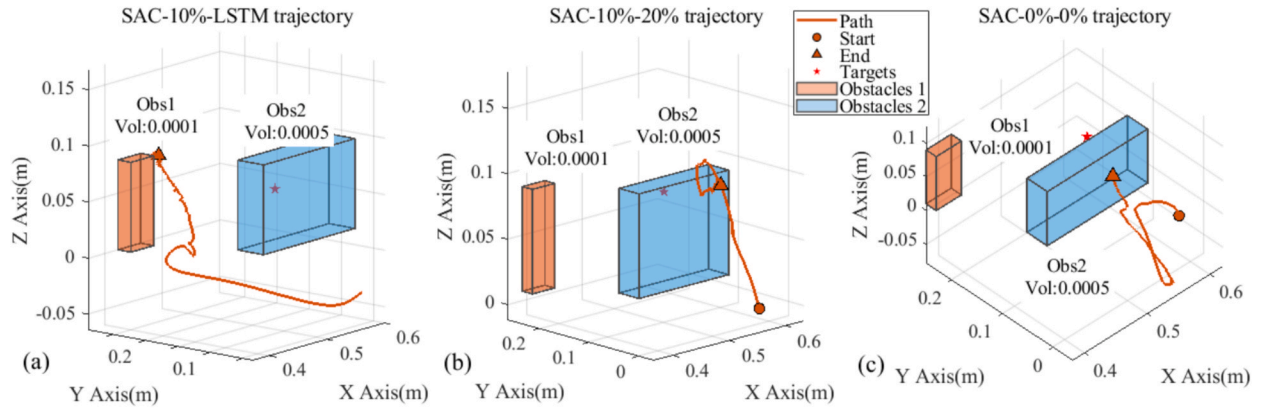


Fig. 11. Agent trajectory without perceptual function.

policy network, which relies solely on instantaneous observations, is unable to construct multi-step plans, resulting in oscillatory behavior and collision with Obs2 during approach. These results demonstrate that, in unstructured and complex environments, the Long Short-Term Memory (LSTM) network effectively enables the agent to perform robust obstacle avoidance by leveraging temporal context and historical experience.

Further analysis indicates that upon the removal of the obstacle-aware reward function in the third stage, the obstacle avoidance task transitions from a reward-driven problem into a typical Partially Observable Markov Decision Process. In this environment, single-step observations and immediate rewards are insufficient to enable the agent to fully explore the feasible space, causing models relying solely on current-state decisions, such as SAC-0%-0%, to exhibit myopic behaviors and fall into local oscillations. In contrast, the SAC-10%-LSTM model incorporates a temporal feature modeling module composed of a two-layer LSTM and a multi-head self-attention mechanism between the observation inputs and the Actor-Critic network. By encoding the history of observations and actions, the two-layer LSTM captures the variation trends of cable tension and the inertial characteristics of joint motion within its hidden states. This enables the policy to make decisions based on motion trends rather than instantaneous states. Even in the absence of the obstacle perception function at the current moment, the hidden state h_t at time t encapsulates trajectory evolution information over the sequence length, allowing the policy network to infer potential collision risks associated with maintaining the current motion trend.

Addressing the joint hysteresis and cable tension coupling effects of the cable-driven manipulator, the LSTM effectively extracts the dynamic rate of change in cable tension and the inertial characteristics of joint motion. This capture of dynamic characteristics ensures that the agent can generate smooth and temporally coherent control commands even without the perception function, avoiding control oscillations caused by myopic decision-making.

The multi-head self-attention mechanism further reinforces selective attention to critical historical states, enabling the network to focus temporally on key time steps associated with obstacle proximity, path deviation, and velocity changes. Consequently, even if the policy receives no reward signals related to obstacles at the current moment, it can still implicitly identify potential high-risk regions through reasoning over historical trajectories and adjust the direction of motion in advance to achieve collision-free avoidance.

The end-effector requires stronger immediate feedback and high-frequency reward signals when approaching the target point. In the absence of such reward constraints, the policy tends to decelerate prematurely and exhibit conservative convergence near the target, resulting in a slight increase in the final arrival distance.

4.5. Hyperparameter analysis of LSTM

Comparative experiments were conducted on three mainstream temporal feature extraction networks (LSTM, GRU, and TCN) while maintaining consistency across the reinforcement learning algorithm framework, observation space definitions, reward function design, and training hyperparameters. The results are presented in the Table 14 and Fig. 12.

Experimental results demonstrate that the SAC-10%-LSTM [Fig. 12 (c)] model exhibits superior comprehensive control performance and robustness. It achieved an average return of 33,400.22, an average arrival distance of 0.0330 m, and an average of 886 decision steps. These data strongly validate that the gated memory mechanism of LSTM effectively captures the significant nonlinear dynamics and hysteresis characteristics of the cable-driven robotic arm system, ensuring policy stability during long-horizon tasks while achieving high-precision convergence.

The overall performance of SAC-10%-GRU [Fig. 12(b)] declined slightly, yielding an average return of 29,945.34 (approximately 10.3% lower than LSTM) and an average final error of 0.0359 m (an increase of approximately 8.8% compared to LSTM). Although the GRU model demonstrated a certain efficiency advantage by reducing the average number of control steps to 667 (a decrease of approximately 24.7% compared to LSTM), its control accuracy and stability were compromised. This further indicates that the simplified gating structure is less effective in capturing long-term dynamic information.

Notably, while the SAC-10%-TCN [Fig. 12(a)] model outperformed LSTM and GRU in specific numerical metrics by achieving the lowest average arrival distance (0.0296 m) and the fewest control steps (226), its average return was extremely low at only 7,290.89. As illustrated in the Fig. 12, this anomaly is attributed to the model frequently colliding with obstacles during both training and testing phases, which resulted in the premature termination of episodes and the triggering of substantial penalties. This indicates that while TCN is capable of rapidly extracting local temporal features and generating aggressive goal-oriented policies, it exhibits distinct myopic behavior when confronted with strongly coupled dynamics and safety constraints.

To investigate the impact of the input sequence length of the LSTM feature extractor on its capability to capture the dynamics of the cable-driven manipulator, a sensitivity analysis was conducted on sequence

Table 14
Agent performance with different temporal models.

Model	Mean Reward	Average Final Dist (m)	Average Step Count
SAC-10%-LSTM	33400.22	0.0330	886
SAC-10%-GRU	29945.34	0.0359	667
SAC-10%-TCN	7290.89	0.0296	226

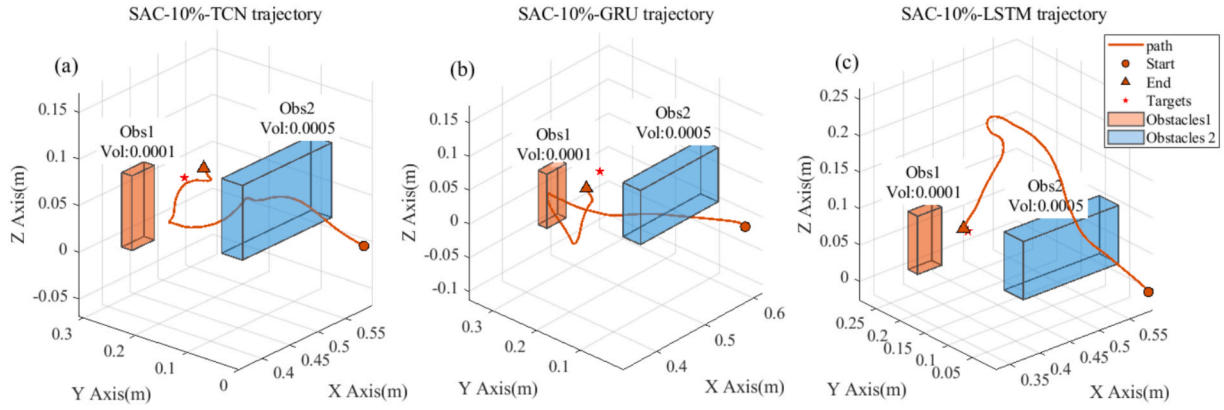


Fig. 12. Agent trajectory with different temporal models.

lengths ranging from 10 to 100, while maintaining other hyper-parameters (such as network depth and hidden layer dimensions) constant. The experimental results, as presented in Table 15, reveal a significant non-linear coupling relationship between sequence length and control performance.

When the input sequence length is insufficient, the agent fails to acquire adequate historical information to reconstruct the hidden states of the manipulator. Specifically, with a sequence length of 10, the model failed to complete the task throughout the training process, yielding a 0% success rate. The final end-effector distance reached 0.1323 m, and the average control steps consistently hit the upper limit (1000 steps), indicating a failure of the policy to converge effectively. Although the success rate improved to 25.0% when the sequence length was increased to 20, overall control performance remained limited, with the model exhibiting a tendency to fall into local optima. In this scenario, the agent tended to rapidly approach the target to maximize distance-related rewards; however, due to insufficient perception of cable hysteresis effects and manipulator inertia, it failed to effectively suppress residual end-effector vibrations, making stable convergence in the vicinity of the target difficult to achieve. This phenomenon demonstrates that excessively short time sequences are inadequate for supporting high-precision control tasks characterized by strong dynamic coupling.

A qualitative leap in model performance was observed when the sequence length was increased to 30. Under this configuration, the task success rate reached 100%, and the average final error decreased to 0.0330 m. With a training time of only 2.00 h, this setting demonstrated an excellent balance between control precision and training efficiency. Furthermore, the model with a sequence length of 30 achieved the optimal cost-benefit ratio while ensuring global convergence.

A sequence length of 70 also achieved a 100% success rate, its training time increased significantly to 5.33 h. In comparison, the training time for a sequence length of 30 was reduced by approximately 62.5%, while the average control steps (886 steps) remained within the

same order of magnitude as those for sequence length 70 (772 steps), further validating the advantage of medium-length sequences in terms of performance and efficiency.

With further increases in sequence length, model performance did not exhibit continuous improvement but rather demonstrated significant instability. Specifically, the success rate dropped to 62.5% for sequence lengths of 40 and 50. For sequence lengths of 60 and 80, the success rate further declined to 50.0% and 37.5%, respectively, accompanied by varying degrees of increase in both final distance and decision steps. This evidence confirms that excessively long input sequences introduce substantial distant historical information that is weakly correlated with current control decisions; this redundant information functions as noise, interfering with the Critic network's accurate estimation of current state-action values.

When the sequence length was extended to 100, the model completely failed to converge, with the success rate dropping to 0% and the average step count returning to the upper limit. This demonstrates the infeasibility of using ultra-long sequences in practical training scenarios.

4.6. Performance evaluation and comparative analysis of LSTM-based agent in complex environments

To further evaluate the generalization capability and robustness of the proposed SAC-10%-LSTM in complex environments, nine test scenarios were constructed based on the third-stage curriculum learning framework, featuring varying numbers (from one to three) and sizes (small, medium, and large) of obstacles. The statistical results of these experiments are summarized in Table 16.

The results demonstrate that the agent maintains strong robustness despite the increased environmental complexity. Notably, the geometric size of obstacles exerts a measurable influence on the path-planning strategy. As the obstacle volume increases, the average number of steps required to complete the task exhibits an overall upward trend. In single-obstacle scenarios, the presence of a large obstacle (Case c) leads

Table 15
Quantitative performance comparison of different sequence lengths.

Sequence Length	Success Rate (%)	Average Final Dist (m)	Average Step Count	Mean Reward	Train Time (h)
10	0	0.1323	1000	1321.12	1.21
20	25.0	0.0728	776	11823.16	1.33
30	100.0	0.0330	886	33400.22	2.00
40	62.5	0.0437	534	17024.90	2.89
50	62.5	0.0485	624	15380.68	3.56
60	50	0.0789	733	11904.06	4.42
70	100.0	0.0347	772	10867.52	5.33
80	37.5	0.0536	798	5671.65	6.47
90	87.5	0.0484	556	7268.17	7.11
100	0	0.1124	1000	541.45	8.19

Table 16
Performance of LSTM-based agent in different obstacle environments.

Case	Obstacle Count	Obstacle Size	Average Step Count	Mean Reward	Average Final Dist (m)
(a)	1	small	384	8408.32	0.031
(b)	1	medium	244	3704.96	0.028
(c)	1	big	746	9838.72	0.040
(d)	2	small	442	17068.11	0.031
(e)	2	medium	886	33400.22	0.033
(f)	2	big	813	26360.77	0.029
(g)	3	small	667	13052.83	0.034
(h)	3	medium	628	6897.89	0.036
(i)	3	big	730	8702.28	0.037

to a substantial increase in the average decision steps to 746, compared with 384 steps for the small obstacle case (Case a). This can be attributed to the fact that larger obstacles occupy a greater portion of the workspace, effectively eliminating potential shortcuts. Furthermore, under multi-obstacle constraints, the model preserves a high level of control accuracy. When the number of obstacles increases to three (Cases g–i), the workspace becomes fragmented; nevertheless, SAC-10%-LSTM consistently achieves an average final distance in the range of 0.028–0.040 m across all test scenarios, satisfying the predefined success threshold of 0.04 m. In Case i, although the average number of steps rises to 730 and the average reward decreases to 8702.28, the agent is still able to successfully reach the target.

Fig. 13 illustrates the obstacle-avoidance trajectories of SAC-10%-LSTM under nine different obstacle configurations. As the geometric complexity of the environment increases progressively, the agent exhibits stable and adaptive obstacle-avoidance behavior. In single-obstacle scenarios [Fig. 13(a)–(c)], the end-effector trajectory gradually transitions from an approximately straight-line path to a curved path with pronounced detouring characteristics. As the obstacle size increases, the agent initiates avoidance maneuvers at an earlier stage and maintains a larger safety margin along the obstacle boundary,

resulting in longer yet smoother detour trajectories.

In dual-obstacle scenarios [Fig. 13(d)–(f)], the feasible workspace becomes distinctly non-convex, substantially increasing the complexity of avoidance decision-making. Nevertheless, the policy consistently generates continuous, collision-free trajectories without frequent backtracking or local oscillations. Notably, in the scenario with two medium-sized obstacles, the trajectory forms a smooth S-shaped path between the obstacles, highlighting the policy's ability to maintain stable control and spatial consistency over long decision horizons.

It is important to note that the increase in trajectory length and average decision steps does not indicate a degradation of the agent's policy. Due to the pronounced tension coupling and response lag inherent in the dynamics of cable-driven manipulators, aggressive path planning near obstacles can easily induce control oscillations or elevate collision risks. By leveraging the LSTM module, the policy network effectively exploits historical state sequences, particularly tension variations and motion trends, to implicitly predict future feasible regions, thereby avoiding potentially high-risk areas.

In high-complexity scenarios involving three obstacles [Fig. 13(g)–(i)], the workspace is further partitioned; however, the agent remains capable of reaching the target through multi-stage detouring strategies.

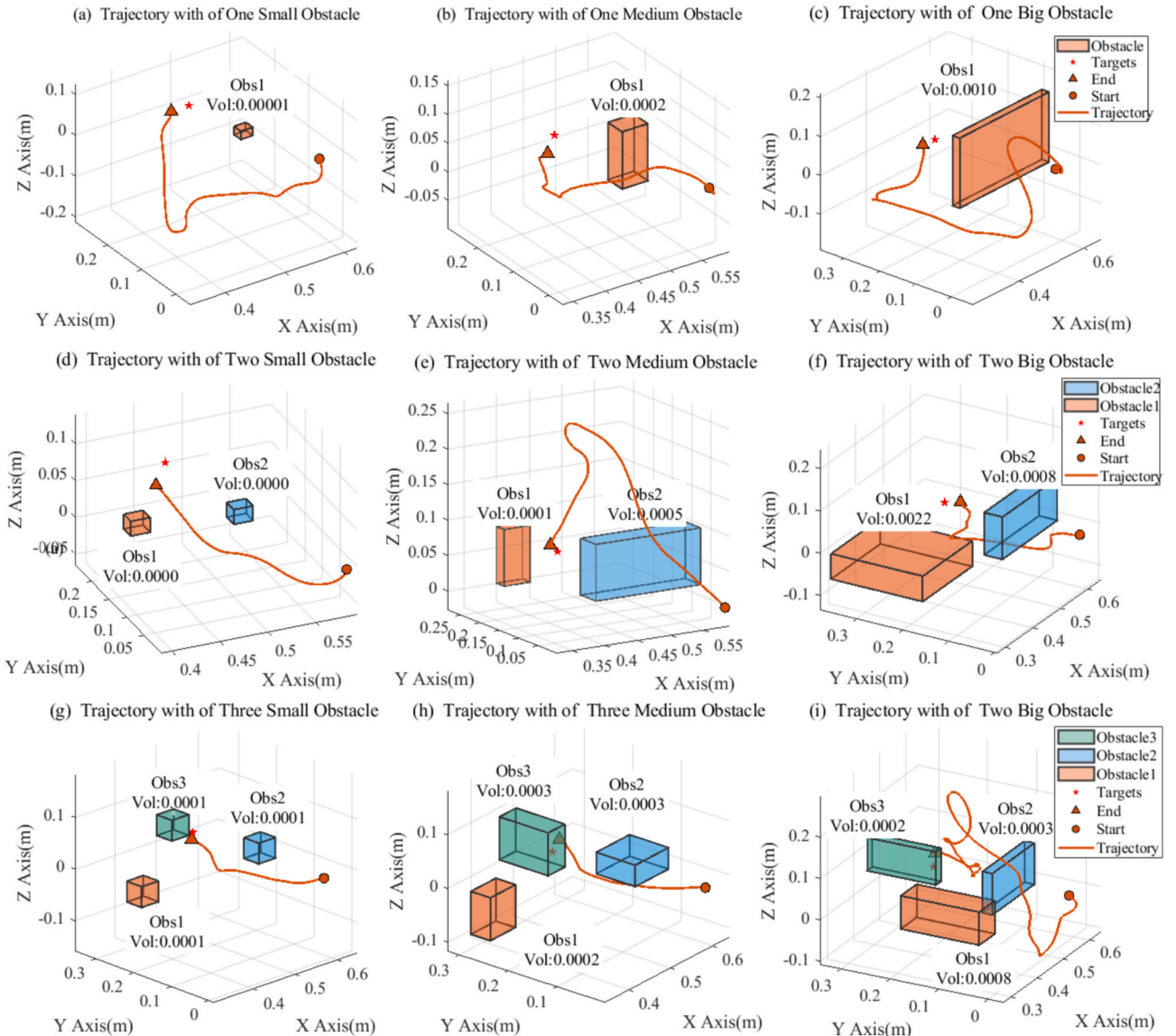


Fig. 13. Agent trajectory in different obstacle environments.

Even in the most challenging scenario with three large obstacles, the trajectories preserve continuity and controllability, demonstrating that the proposed method achieves strong generalization capability and robustness in complex environments.

As shown in Table 17, the proposed SAC-10%-LSTM method demonstrates comprehensive performance superior to existing state-of-the-art methods when applied to Cable-Driven Redundant Manipulators characterized by strong nonlinearity and hysteresis. Although the incorporation of LSTM results in a relatively longer training time, its ability to effectively memorize historical states enables the achievement of collision-free motion planning in complex environments with dynamic obstacles. In contrast, baseline methods exhibit shortcomings in robustness: Shen et al. (Shen et al., 2022) suffer from oscillations at constraint boundaries due to rigid constraints, resulting in a success rate of only 96.8%; Huang et al. (Huang et al., 2023) experience approximately 10% geometric deadlocks caused by strict null-space constraints; regarding Ma et al. (Ma et al., 2025), which relies on offline manipulation tasks based on static datasets, the experimental results are not directly comparable (marked as N/A), as this work aims to address reactive obstacle avoidance in dynamic scenarios.

The reproduction experiment regarding the SAC-APF method proposed by Bai et al. (Bai et al., 2023) reveals significant limitations (as shown in Fig. 14(b)). The SAC-APF [Fig. 14 (b)] trajectory on the right indicates that this approach is critically affected by the local minima issues intrinsic to the Artificial Potential Field (APF). In the scenario of a narrow passage with dual obstacles (Obs1 and Obs2), the mutual cancellation between the target's attractive potential and the obstacles' repulsive potentials causes the manipulator to fall into a potential trap and stall prematurely. These results indicate that, while the proposed method sacrifices some training efficiency, this architectural improvement is justified for cable-driven systems to resolve hysteresis effects, eliminate oscillations.

4.7. Future outlook

4.7.1. Deployment feasibility analysis

A preliminary analysis was conducted in this study. The proposed end-to-end SAC-LSTM control strategy demonstrates a single inference latency of approximately 8–10 ms (GPU accelerated) on the NVIDIA Jetson Orin edge computing platform. When combined with sensor data acquisition and communication overhead, the overall control loop latency can be kept within 20 ms, meeting the requirement for a 30 Hz real-time control frequency.

The policy network contains approximately 500,000 parameters, and the model file size is about 2.45 MB, making it suitable for deployment on resource-constrained embedded platforms. Future work will further compress the model size using lightweight techniques such as knowledge distillation and conduct systematic latency testing and power consumption evaluations on actual hardware platforms to verify the algorithm's feasibility in unstructured application scenarios.

4.7.2. Sim-to-real transfer and dynamic compensation strategy

To address the issue of transferring the simulation strategy to real robot control, this study employs multi-level technical measures to bridge the gap between simulation and the physical machine, as shown

in Fig. 15.

The reinforcement learning control node serves as the core of the system, subscribing to sensor topics and executing forward inference of the PyTorch policy network. The model outputs a 12-dimensional normalized action vector representing the target speed for each motor. An action post-processing module applies Exponential Moving Average (EMA) smoothing to the raw output to suppress high-frequency jitter. Subsequently, a denormalization operation is performed to map the actions to the actual motor speed range, followed by the application of velocity and acceleration constraints. The processed control commands are published via ROS topics to the serial driver node, encapsulated into data frames according to the communication protocol, and sent to the lower-level controller. The lower-level controller then parses these commands to drive each motor to the corresponding speed.

Notably, during the initial model training phase, this study configured key dynamic parameters, such as motor torque constants and damping coefficients, in the MJCF model file based on the physical parameters of the actual motors. This ensures that the simulation environment accurately reflects the dynamic characteristics of the real system. Furthermore, to enhance the policy's robustness against parameter uncertainty, domain randomization strategies were introduced during the training phase, applying random perturbations to joint friction coefficients, damping parameters, sensor noise, and control latency.

5. Conclusion

To address the complexities of high-dimensional control and non-linear hysteresis in simulated Cable-Driven Redundant Manipulators (CDRMs), a curriculum learning framework incorporating dynamic expert guidance and LSTM-based temporal modeling was developed. By integrating these mechanisms, the proposed method not only demonstrates superior navigation in sparse-reward environments but also substantially enhances the robustness of motion planning within virtual simulations. The key findings of the research are summarized below:

Through a comparative evaluation of expert strategies, the Artificial Potential Field (APF) method was identified as the most efficient guidance source, reducing trajectory generation time to 30 min while maintaining optimal path quality. Building upon the APF guidance, a dynamic hybrid-ratio strategy (5%–20%) was implemented. Simulation results indicate that dynamically adjusting the expert intervention effectively mitigates policy locking, yielding an approximate 89% improvement in positioning accuracy (reaching 0.029 m) compared to the pure reinforcement learning baseline.

Furthermore, the LSTM-integrated architecture successfully resolved the dynamic coupling issues inherent in the modeled CDRM. In ablation studies, the agent utilizing an optimal sequence length of 30 achieved a 100% success rate and a mean reward of 33,400, significantly outperforming GRU and TCN baselines. Notably, the framework maintained collision-free navigation even in simulated scenarios lacking explicit obstacle-aware rewards, a capability that validates the model's ability to infer risks directly from historical motion trends.

These findings underscore the potential of combining structured expert knowledge with temporal deep learning to optimize complex non-linear systems. Although the current conclusions are strictly derived

Table 17
Performance comparison with SOTA motion planning methods.

Dimension	SAC-10%-LSTM	Shen et al. (Shen et al., 2022)	Huang et al (Huang et al., 2023)	Ma et al (Ma et al., 2025)	Bai et al (Bai et al., 2023)
Target System	Cable-Driven Redundant Manipulator	Rigid Multi-Joint Arm	General Redundant Arm	Space Robot	General Manipulator
Success Rate	100% (3 Dynamic Obstacles)	96.8% (2 Static Obstacles)	91.4% (1–2 Obstacles)	N/A	N/A
Network Arch	LSTM-MLP	MLP	MLP	MLP	MLP
Key Limitation	long training time	Rigid constraints cause jitter	10% failure due to strict constraints	Fails on unseen states/obstacles	Trapped by APF potentials

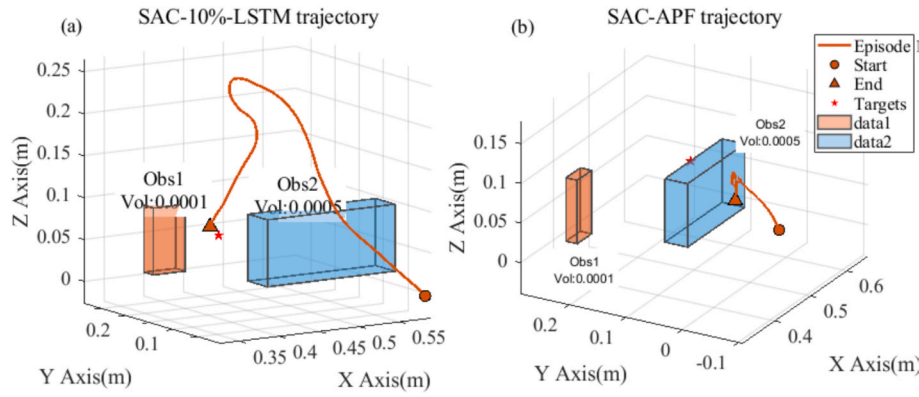


Fig. 14. Agent trajectory comparison in medium-obstacle environments.

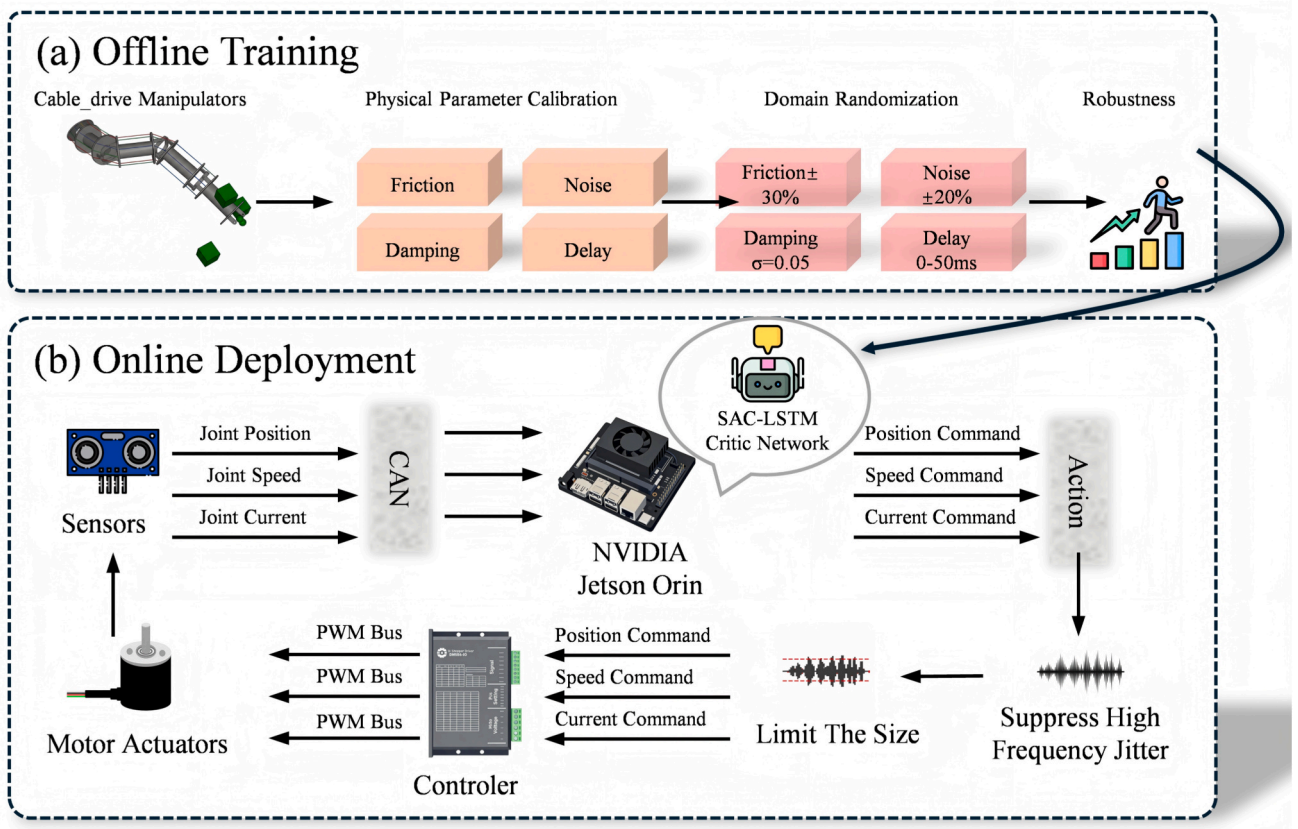


Fig. 15. Framework of sim-to-real transfer and control deployment.

from simulation data, the successful integration of data-driven approaches establishes a validated theoretical foundation for future sim-to-real transfer applications.

CRediT authorship contribution statement

Biya Cheng: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Resources, Data curation, Writing - Original Draft, Writing - Review & Editing, Visualization, Supervision, Project administration, Funding acquisition. **Xinde Zhang:** Investigation, Software, Formal analysis, Data curation, Writing - Original Draft. **Chuang Ma:** Investigation, Software. **Jiaxiang Li:** Investigation, Software. **Kaixiang Huang:** Investigation, Data curation. **Chiliang Zhong:** Investigation, Data curation. **Yingyuan Guan:** Investigation, Data curation. **Xueming Yin:** Investigation, Software. **Yuyuan Qiu:** Formal

analysis, Investigation. **Xuan Wang:** Investigation, Software. **Hongjun Wang:** Supervision, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

The source code of this study has been made publicly available and can be accessed via the repository link (https://github.com/emmawithsai-lab/Cable_driven_RL) provided in the manuscript.

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